

**Treatment Success and Failure Prediction Using Machine Learning Models
for Pulmonary Tuberculosis Patients Under the Department of Health-Cordillera
Administrative Region (DOH-CAR) Tuberculosis-Directly Observed Treatment,
Short-course (TB-DOTS) Program**

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Abstract

Pulmonary Tuberculosis (PTB) remains a critical public health challenge in the Cordillera Administrative Region (CAR), Philippines, where complex patient profiles and programmatic gaps contribute to persistent treatment failure. This study develops machine learning models to predict treatment success or failure among PTB patients in DOH-CAR TB-DOTS facilities covering the period 2015–2025. A standardized preprocessing pipeline performs schema harmonization, patient pseudonymization, and a diagnostic-first missing data handling protocol that classifies each variable's missingness mechanism—Missing Completely at Random (MCAR), Missing at Random (MAR), or Missing Not at Random (MNAR)—before routing it to one of four targeted strategies: column exclusion, listwise deletion, missing indicator imputation, or stochastic Multivariate Imputation by Chained Equations (MICE). Supervised models—Logistic Regression, Support Vector Machines (SVM), and Extreme Gradient Boosting (XGBoost)—are trained alongside Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) temporal architectures to capture longitudinal treatment dynamics across the 6- to 12-month treatment course. Shapley Additive Explanations (SHAP) ensure model interpretability by quantifying the contribution of individual patient features to each prediction. The validated models are embedded in a Clinical Decision Support System (CDSS) that provides dynamic risk estimates and factor-level explanations to support early clinical intervention for patients at risk of treatment failure.

Keywords: tuberculosis, machine learning, treatment outcome prediction, clinical decision support system, explainable AI

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Pulmonary Tuberculosis (PTB) is a chronic infectious disease caused by *Mycobacterium tuberculosis* (MTB), a slow-growing and acid-fast bacillus that continues to challenge public health due to its gradual progression and risk of treatment failure (Tobin & Tristram, 2024). Despite effective therapies, PTB remains widespread due to its strong association with poverty, undernutrition, and comorbidities such as diabetes and HIV which weaken immune defenses and increase susceptibility (Tobin & Tristram, 2024). These factors delay diagnosis, complicate treatment, and contribute to disease progression. Globally, TB is still one of the leading causes of death, with 1.5 million deaths and 10 million new cases recorded in 2018, including half a million with rifampicin resistance.

In the Philippines, TB continues to be a major public health concern. The World Health Organization (WHO) identifies the country as one of the top contributors to the global TB burden as it accounts for an estimated 6.8 percent of cases. Before the COVID-19 pandemic, the Philippines had one of the highest TB rates in Asia with over 554 cases per 100,000 people in 2019 (Florentino et al., 2022). In 2023, around 37,000 Filipinos died from TB out of the 739,000 diagnosed cases. The WHO classifies the country as a high burden for MDR-TB and TB/HIV co-infection (The Borgen Project, 2025). That same year, the DOH recorded 575,770 TB cases through its Integrated TB Information System which indicates that many cases remain unregistered or undiagnosed (Baron, 2025). The DOH implements CDC-recommended regimens for active and latent TB through the TB-DOTS strategy which was expanded in 2003 to include private clinics, pharmacies, and hospitals (Bendre et al., 2021; Department of Health [DOH], 2014). Despite these initiatives, gaps in diagnosis, treatment monitoring, and patient support persist especially in the management of MDR-TB (Stop TB Partnership, 2021).

In Baguio City, the DOH-CAR reported 437 confirmed TB cases in 2022 along with

over 4,405 missing or unreported cases making the area a priority for surveillance and treatment improvement (Department of Health - Republic of the Philippines, 2022). Contributing factors include high population density, internal migration, climate, and the city's role as a referral center for nearby provinces. Patient comorbidities, socioeconomic barriers, and health system limitations also influence treatment outcomes (Khalid et al., 2022). These challenges highlight the need for locally adaptive strategies as treatment failure not only worsens patient outcomes but also prolongs transmission and drives the emergence of MDR-TB (World Health Organization, 2022).

Risk Factors for Treatment Failure

To ensure uniformity in the evaluation of TB control efforts, the DOH has established standardized criteria for classifying treatment outcomes through the National Tuberculosis Program (NTP). Establishing these outcomes is crucial for monitoring program performance, ensuring comparability across studies, and informing risk factor analysis. Ultimately, these outcomes of success or failure are not random but are linked to a distinct set of patient-specific, clinical, and programmatic risk factors. While mortality remains the primary outcome for most studies on treatment failure, some also consider the development of disease, resistance to multiple anti-TB drugs, and failure to follow up (Department of Health [DOH], 2014).

One of the risk factors is that TB patients have close contact with other patients with active TB infection. A study noted that TB patients who acquired the infection through close contact were shown to be at higher risk of delayed diagnosis and more likely to be non-adherent with the treatment either by refusing or discontinuing the regimen (Lamsal et al., 2009). Another risk factor is recurrence of the infection or having a history of receiving TB treatment. TB patients who missed follow-ups had a higher risk of reactivation of the infection and mortality as they are more difficult to treat compared to patients with no prior history of TB (Tok et al., 2020). Furthermore, missed or interrupted treatment causes the development of resistance to the drug if done frequently (Ndambuki et al., 2020).

Comorbidities are also discussed in studies that have consistently shown a significant influence on tuberculosis treatment outcomes across different settings. In Eastern Ethiopia, HIV co-infection was determined to be a substantial risk factor for treatment failure due to the weakened immune system which leads to a higher risk for ineffective treatment (Amante & Abdosh, 2015). In addition, TB and HIV drug interactions accelerate each other's progression and increase the occurrence of side effects or adverse effects which may compromise treatment success (Azeez et al., 2018). Likewise, a study in Davao City, Philippines, revealed that there was a higher risk of treatment failure in comorbid patients with 58.82% of the 51 patients who experienced treatment failure having at least one comorbidity (Hinay et al., 2025). Another major determinant is Diabetes Mellitus (DM). TB patients with diagnosed DM have a higher risk of treatment failure, relapse, and death compared to non-diabetics (Noubiap et al., 2019). In a systematic review and meta-analysis, patients with DM have a higher risk of experiencing treatment failure for Drug-resistant (DR) and mortality for MDR-TB with an odds ratio of 1.56 and 1.57 respectively (Xu et al., 2023). In line with this, another study states that patients suffering from both PTB and chronic conditions, particularly DM, had reduced rates of sputum conversion with a higher probability of poor treatment outcomes including death and progression to MDR-TB (Siddiqui et al., 2016).

Taken together, these findings reveal that TB treatment failure is driven not by any single factor but by the complex interplay of comorbidities, programmatic gaps, and socioeconomic vulnerabilities that conventional single-factor analyses fail to capture, underscoring the need for an integrative machine learning framework capable of modeling these multidimensional interactions to enable earlier and more targeted clinical intervention (Amante & Abdosh, 2015; Department of Health - Republic of the Philippines, 2022; Hinay et al., 2025; Khalid et al., 2022; Noubiap et al., 2019; Peng et al., 2024; Rajpurkar et al., 2022; World Health Organization, 2022; Xu et al., 2023).

Machine Learning for TB Outcome Prediction

Although multiple studies have examined risk factors such as sociodemographic characteristics, comorbidities, and medical history, the findings remain inconsistent and fragmented since methodological constraints limit them. Most studies rely on traditional statistical analyses which are statistical tests or regressions that examine one factor at a time or a few factors together while assuming their linear relationships (Amante & Abdosh, 2015). These limitations highlight the need for an integrative approach to combine multiple dimensions of data in analyzing the interrelated variables together as a critical gap in capturing the multifaceted nature of TB treatment outcomes.

To address this gap, the present study adopts a holistic multi-layer analytical framework that integrates descriptive analysis and supervised predictive modeling using demographic, clinical, diagnostic, and treatment-related variables. Descriptive analyses are employed to characterize patient profiles and treatment patterns, while supervised learning models are used to predict treatment outcomes. This integrative approach enables machine learning models to learn complex relationships among multiple patient-level variables that influence treatment success or failure, thereby enhancing both predictive accuracy and interpretability (Shickel et al., 2018; Smith et al., 2021).

Machine learning (ML) offers a distinct advantage in modeling complex and non-linear interactions among demographic, clinical, diagnostic, and treatment-related variables that conventional statistical analyses may overlook. By leveraging algorithms capable of learning from high-dimensional data, such as logistic regression, Support Vector Machines (SVM), and ensemble-based methods including Extreme Gradient Boosting (XGBoost), the study aims to uncover subtle dependencies that contribute to treatment success or failure (Rajpurkar et al., 2022). Previous research has demonstrated the utility of ML in TB research, including predicting drug resistance, identifying high-risk patients, and optimizing treatment strategies (Kaur & Sharma, 2021; Kosakov et al., 2024). In this study, the models are trained directly on integrated patient data, allowing them to capture individual-level patterns that improve predictive performance while maintaining interpretability.

Beyond prediction, interpretability will be enhanced through the application of Shapley Additive Explanations (SHAP) which is a post-hoc interpretability technique that quantifies the contribution of each variable to the model's predictions. SHAP enables a transparent evaluation of which patient features most strongly drive treatment success or failure. This supports explainable and trustworthy Artificial Intelligence (AI) in clinical contexts (Peng et al., 2024). Using supervised learning with interpretable AI methods, this framework not only forecasts treatment outcomes but also reveals the relative importance of risk factors such as drug resistance, comorbidities, or regimen adherence. The resulting models are designed to help healthcare providers identify patients at a higher risk of treatment failure thus enabling early intervention, tailored regimens, and improved patient management.

Contributions

The main contributions of this paper are as follows:

1. **Integrated Predictive Framework.** The team develop and validate machine learning models that accurately predict therapy success or failure among TB patients in the CAR, Philippines. The integration of demographic, clinical, and laboratory variables focuses on supporting early identification of patients at risk of unsuccessful treatment and providing evidence-based tools for improving TB program outcomes.
2. **Data Preprocessing and Harmonization.** The team pre-process and document patient records of PTB cases from TB-DOTS facilities in the CAR through data cleaning, feature engineering, and reporting for readiness to be used in ML applications.
3. **Descriptive and Exploratory Analysis.** The team analyzed the demographic, clinical, diagnostic, and treatment characteristics of PTB patients to provide a comprehensive overview of the study population and identify key variables associated with treatment outcomes over the course of therapy.
4. **Temporal Modeling.** The team incorporates temporal modeling that analyzes time-

dependent patterns in patient data allowing the detection of dynamic changes and trends throughout treatment. This thereby improves the accuracy of outcome predictions and supports timely clinical interventions.

5. **Clinical Decision Support.** The team developed a decision support tool that visualizes patient risk levels and predictive outputs. This enables early identification of individuals at risk of poor outcomes and ensures that the model maintains high predictive accuracy and reliability when applied to evaluate incoming TB cases.

Research Objectives

This study aims to develop and validate machine learning models that accurately predict therapy success or failure among TB patients in the CAR, Philippines. The integration of demographic, clinical, and laboratory variables focuses on supporting early identification of patients at risk of unsuccessful treatment and providing evidence-based tools for improving TB program outcomes, with the overarching goal of enabling personalized clinical interventions and efficient allocation of healthcare resources. Specifically, the research aims to achieve the following objectives:

1. To pre-process and document patient records of PTB cases from TB-DOTS facilities in the CAR through data cleaning, feature engineering, and reporting for readiness to be used in ML applications.
2. To describe the demographic, clinical, diagnostic, and treatment characteristics of PTB patients in the dataset, providing an overview of the study population.
3. To identify contributing variables that influence treatment outcomes over the course of therapy, enabling recognition of the most critical factors that affect patient recovery and guiding more precise treatment adjustments that can be intervened with.
4. To incorporate temporal modeling that analyzes time-dependent patterns in patient data, allowing the detection of dynamic changes and trends throughout treatment, thereby improving the accuracy of outcome predictions and supporting timely clinical interventions.

5. To train and validate supervised learning models for predicting treatment success or failure among PTB patients, enabling early identification of individuals at risk of poor outcomes and ensuring that the model maintains high predictive accuracy and reliability when applied to evaluate incoming TB cases.
6. To develop a decision support tool that visualizes patient risk levels and predictive outputs.

Scope and Limitations

The dataset will include PTB patients for the period 2015–2025 in the CAR excluding those who are still undergoing treatment at the time of data extraction. Furthermore, only patients with a well-defined treatment outcome of either success or failure will be included. The data gathered will encompass demographic information, laboratory tests, diagnostic data, clinical data, treatment regimen, follow-up examinations, drug administration data, and treatment outcomes. The study faces limitations related to the sufficiency and quality of the dataset. The data sourced from existing records may present challenges related to completeness, consistency, and representativeness due to limited access to complete clinical parameters. This could affect the inclusion of important risk factors and weaken the robustness of the analysis. Additionally, the study may not fully represent the current TB patient population as patients currently undergoing treatment are excluded. Difficulties are also expected during data extraction from DOH-CAR as variations in record-keeping over the 10-year period from 2015 to 2025 may be inconsistent due to changes in the healthcare system and shifts in documentation standards. Reliance on self-reported indicators may also introduce bias and inconsistencies that could affect the predictive capabilities of the model.

Significance of the Study

The resulting models are designed to help healthcare providers identify patients at a higher risk of treatment failure thus enabling early intervention, tailored regimens, and improved pa-

tient management. Ultimately, this machine learning framework supports the broader objective of strengthening TB control efforts in the Cordillera Administrative Region (CAR) by transforming existing patient data into predictive, explainable, and actionable tools for evidence-based public health practice. The results will help improve patient management and treatment planning, strengthen program implementation, and guide public health decision-making to improve the TB treatment success rate in CAR. Furthermore, the study aims to maximize benefits for the public by informing health administrators about the model which may be further developed into a solution or application to aid the local TB-DOTS program. The processed and pseudonymized derivative of the DOH-CAR PTB Treatment Outcome Dataset produced by the study will also be published to support the advancement of clinical, epidemiological, and machine learning research.

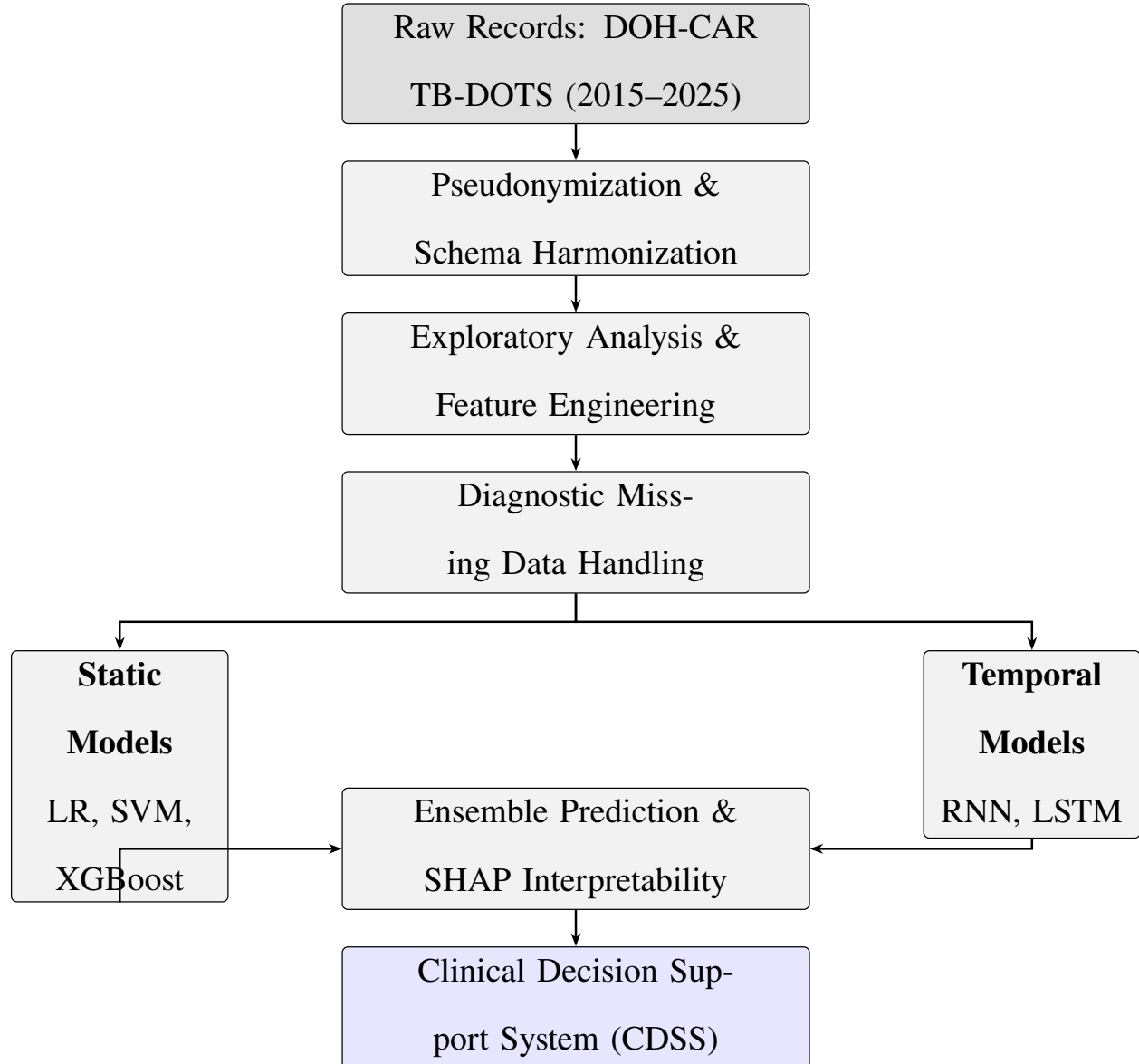
Method

This study uses a dataset of records of PTB patients from the Cordillera Administrative Region (CAR), Philippines, covering 2015–2025 and consolidated by the Department of Health–CAR (DOH-CAR). After ethical approval, records are pseudonymized and processed using a standardized preprocessing pipeline, including schema harmonization, error correction, and feature scaling, followed by a diagnostic-first missing data handling protocol described in the Missing Data Handling subsection below. Exploratory and cluster analyses are conducted to characterize the cohort and derive subgroup-level features. Supervised models, Logistic Regression, Support Vector Machines (SVM), and Extreme Gradient Boosting (XGBoost), are trained to predict treatment outcomes defined by National Tuberculosis Program criteria. To capture longitudinal treatment dynamics, temporal models using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) are evaluated alongside static models, with ensemble strategies applied for improved robustness. Model performance is assessed using standard classification metrics, and Shapley Additive Explanations (SHAP) are employed to ensure interpretability. The final models are integrated into a clinical decision support framework to provide dynamic risk estimates throughout the treatment

course. An overview can be found in Figure 1.

Figure 1

Data Management and Analysis Process



Data Acquisition and Preprocessing

The study utilizes a retrospective dataset of PTB patients from the Cordillera Administrative Region (CAR), Philippines, covering the period from 2015 to 2025. Data is centrally consolidated

by the Department of Health-CAR (DOH-CAR), which manages reports from government TB-DOTS facilities and private practitioners under the Public-Private Mix DOTS (PPMD) scheme (Department of Health [DOH], 2014).

The raw dataset comprises demographic variables (e.g., age, sex, civil status), clinical indicators (e.g., BMI, blood pressure), and diagnostic results (e.g., GeneXpert, Smear Microscopy). To ensure data quality for machine learning applications, a reproducible preprocessing pipeline is implemented. This includes schema harmonization to standardize variable formats and units, and the removal of direct identifiers to strictly enforce patient privacy. Missing data, a common challenge in clinical registries, is addressed through a diagnostic-first protocol described in the Missing Data Handling subsection below.

Exploratory Data Analysis

Prior to predictive modeling, an exploratory analysis was conducted to characterize the study population, assess data completeness, and quantify the associations among variables to inform downstream preprocessing and modeling decisions.

Table 1

Missing Data Summary by Column

Column	Missing_Count	Missing_Percentage
Days_To_Microscopy_Result	7297	98.75
Days_To_RDT_Result	6638	89.84
Days_Screening_To_Diagnosis	4444	60.14
Days_To_Treatment	4378	59.25
Diagnosis_Season	3953	53.5
Diagnosis_DayOfYear	3953	53.5

Table 1 (continued)

Column	Missing_Count	Missing_Percentage
Diagnosis_Quarter	3953	53.5
Diagnosis_Month	3953	53.5
Microscopy Result	2792	37.79
Brgy	1026	13.89
City/Municipality	105	1.42
Province	71	0.96
Treatment Health Facility	29	0.39
Region	15	0.2
Date of Outcome/Status	2	0.03
Bacteriologic Status	1	0.01
Type	0	0
Validation Status	0	0
Age	0	0
Sex	0	0
Anatomical Site	0	0
Source of Patient	0	0
Screening/Diagnosing Health Facility	0	0
RDT Result	0	0
Year	0	0
Outcome/Status	0	0
Registration Group	0	0
Computed_Age	0	0

A global missingness audit was performed across all 28 columns to identify the sparsity structure of the dataset (Table 1). Per-column missing rates were ranked, revealing that variables such as microscopy result (37.8% missing), diagnosis-derived temporal features (53.6%), and clinical delay intervals (59.3–

98.8%) exhibited substantial missingness. This audit provided the empirical basis for the missingness thresholds applied in the Missing Data Handling subsection.

Pairwise correlations among continuous variables were computed using Pearson's r on available-case observations. The numerical set included age, four derived clinical delay intervals (days from screening to diagnosis, days to treatment, days to microscopy result, and days to RDT result), and diagnosis day-of-year. The purpose was to determine whether multivariate imputation via MICE was warranted: if continuous predictors carry meaningful correlation structure, MICE can borrow signal from observed features to estimate missing values; if uncorrelated, univariate mean-fill would be equally effective.

For categorical variables, pairwise association strength was quantified using Cramér's V , a chi-square-based measure normalized to the $[0, 1]$ interval. Columns with fewer than two unique non-null values or fewer than ten non-null observations were excluded from the matrix. The categorical set examined included patient demographics (sex, type of TB, anatomical site), clinical classifications (registration group, bacteriologic status, microscopy result, RDT result), geographic identifiers (province, region), facility-level attributes (screening and treatment health facility), source of patient, and treatment outcome. This analysis detected inflated associations arising from high-cardinality columns and confirmed that the categorical feature space carried sufficient structure for predictive modeling.

Additionally, the study employs cluster analysis on demographic, clinical, and diagnostic dimensions to detect natural groupings of patients. These clusters reveal distinguishing features such as specific comorbidity profiles, adherence behaviors, and regimen responses. Importantly, the resulting cluster assignments are not merely descriptive; they are integrated as engineered features into the subsequent supervised learning models. This approach enables the algorithms to learn both individual-level attributes and subgroup-level patterns that significantly influence treatment success or failure.

To assess data quality of the temporal monitoring dataset prior to preprocessing, a global missingness audit was conducted across all features. Missing rates were computed per column and ranked in descending order. Features exceeding 80% missingness were flagged for the column exclusion pathway, while features between 50% and 80% missingness were flagged for the missing indicator with fill pathway, contingent on predictive importance. This threshold-based triage follows established missing data handling frameworks and ensures that handling strategies are matched to the severity of missingness rather than applied uniformly (Emmanuel et al., 2021).

Patient retention across the temporal dataset’s treatment months was analyzed by computing the proportion of patients with non-missing adherence data at each monthly interval from M0 through M12. Because retention declines naturally over a six- to twelve-month treatment course, a simple per-month count is insufficient; instead, the decay in available observations was characterized as a percentage of the initial cohort at each successive month. This analysis directly informs the temporal modeling scope: months beyond which fewer than 25% of patients retained data are excluded from feature-level correlation analyses to avoid estimates driven by a non-representative surviving subset.

Pairwise correlations among vital signs in the temporal monitoring dataset were computed using Pearson’s r on available-case observations. The presence and strength of inter-vital correlations directly supports the choice of multivariate imputation over univariate strategies: if vital signs are meaningfully correlated, a multivariate method such as Multiple Imputation by Chained Equations (MICE) can borrow signal from observed features to estimate missing values; if uncorrelated, MICE offers no advantage over mean- or median-fill (Aracri et al., 2025). The vital set examined included age, heart rate, oxygen saturation, systolic blood pressure, diastolic blood pressure, weight, height, respiratory rate, and temperature, where present in the temporal records.

To examine how temporal treatment features relate to each other, Pearson correlations were computed on per-patient observed means across Months 0 through 7. For each temporal feature — including dose adherence metrics (monthly doses taken, cumulative doses taken, percentage adherence), anthropometric measurements (weight, height), and diagnostic results (smear microscopy, Xpert MTB/RIF) — the observed values across a patient’s available monthly visits were averaged. Retention characteristics (last month attended, total observed months) were derived from the same window. This approach preserves the dropout signal: a patient who stops attending after M4 contributes means based only on M0–M4, and their limited observation window itself becomes informative rather than requiring imputation.

For categorical variables within the temporal dataset, pairwise association strength was quantified using Cramér’s V , a chi-square-based measure normalized to the $[0, 1]$ interval. A mixed Pearson-Cramér’s matrix was not used, as Pearson’s r assumes interval-level measurement and produces misleading values when applied to nominal or ordinal categories with unequal category spacing (Kossakov et al., 2024). Computing Cramér’s V separately on categorical features avoids this assumption violation and produces interpretable association estimates. Columns with fewer than two unique non-null values or fewer than ten

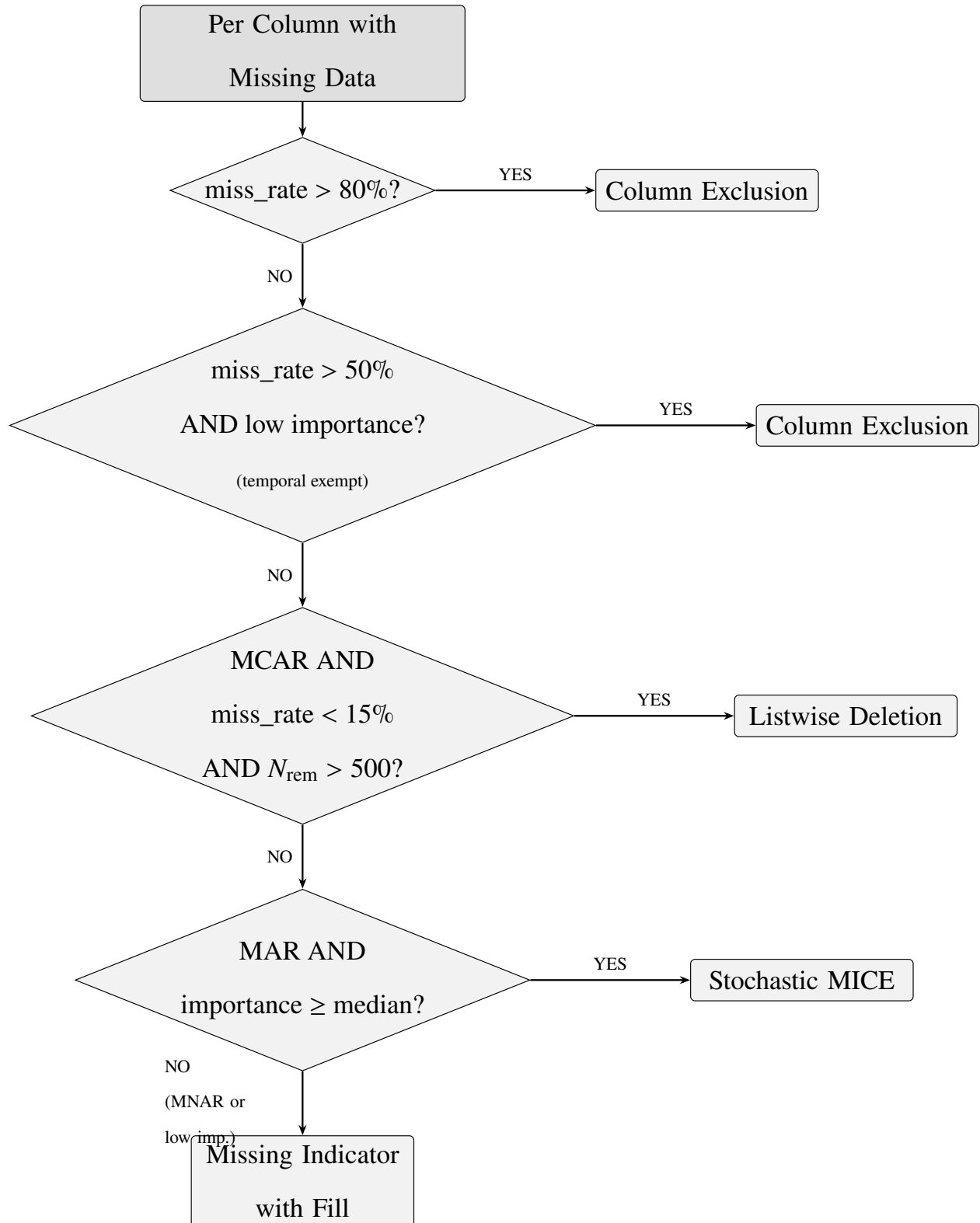
non-null observations were excluded from the matrix, as association cannot be meaningfully estimated under those conditions.

To identify the extent of inconsistent coding that necessitated categorical harmonization, a categorical cardinality audit was conducted across all categorical features in the raw dataset. For each feature, the number of unique raw values was counted to quantify the degree of entry inconsistency. Features with cardinality far exceeding their expected number of valid categories — such as clinical observation fields that should contain a small set of standardized codes but instead contained hundreds of free-text variants — were flagged for harmonization in Stage 4 of the preprocessing pipeline. This audit directly motivates the harmonization targets by providing an empirical baseline of the data entry inconsistency problem before any cleaning was applied.

Finally, the temporal stability of adherence behavior across the treatment course was examined through lag-1 Pearson correlations of percentage adherence across consecutive month pairs (M0–M1 through M7–M8). Month-to-month correlation strength determines whether a recurrent architecture, which assumes sequential dependence between time steps, is warranted. If consecutive months are weakly correlated, a static model operating on aggregated patient means would be equally informative; if strongly correlated, the temporal structure justifies the additional complexity of an RNN or LSTM (Laine & Aila, 2016; Nayebi et al., 2022).

Missing Data Handling

Clinical TB registries exhibit heterogeneous missingness arising from inconsistent documentation, varying facility capacities, and the longitudinal structure of treatment monitoring. Applying a single imputation strategy uniformly across all variables is statistically inappropriate when the mechanism generating missingness differs across features. Accordingly, this study implements a diagnostic-first missing data pipeline that first classifies each variable’s missingness mechanism and then routes the variable to one of four handling strategies: column exclusion, listwise deletion, missing indicator with fill, or stochastic Multiple Imputation by Chained Equations (MICE) imputation (Figure 2) (Dong & Peng, 2013; Emmanuel et al., 2021).

**Figure 2***Missing Data Handling Decision Logic*

The first diagnostic step applies Little’s MCAR test at the dataset level to evaluate whether data are Missing Completely at Random (MCAR). Under the null hypothesis of MCAR, missingness is independent of all observed and unobserved values, meaning that complete cases constitute an unbiased subsample of the population and listwise deletion would be permissible. The test returned a statistically significant result ($p < 0.05$), rejecting MCAR and confirming that missingness patterns in the dataset are systematically related to the observed data, thus ruling out blanket listwise deletion (Dong & Peng, 2013).

Because Little’s test yields a single omnibus verdict at the dataset level, per-column mechanism classification requires a separate procedure. Following the dataset-level assessment, each variable’s missingness indicator is examined against the observed data to assign a column-level mechanism. For temporal (monthly) variables, Spearman rank correlations are computed between the column’s monthly missingness rate and treatment month; a monotonically increasing correlation indicates a dropout pattern classified as Missing Not at Random (MNAR). For static baseline variables, point-biserial correlations between the binary is-missing indicator and observed features are evaluated; significant associations suggest that missingness is explained by other observed variables (Missing at Random, or MAR), whereas non-significant associations indicate MNAR. This per-column classification, rather than a single dataset-level verdict, allows the pipeline to match each handling strategy to the specific statistical properties of each variable (Dong & Peng, 2013; Emmanuel et al., 2021).

To support the column exclusion and MICE eligibility decisions, a Random Forest classifier is trained on available cases, defined as rows where the treatment outcome is observed. Features are filled with column medians for ranking purposes only (available-case analysis), providing a coarse but sufficient signal for importance estimation. Variables in the bottom quartile of predictive importance become eligible for soft column exclusion, while variables above the median importance threshold are prioritized for stochastic imputation (Aracri et al., 2025).

Variables exceeding 80% missingness are excluded entirely under a hard threshold, regardless of their importance ranking, because reliable imputation is statistically untenable at such sparsity levels. A soft threshold additionally excludes variables with more than 50% missingness whose importance falls in the bottom quartile, on the grounds that the information gain from imputing a low-value sparse variable does not justify the synthetic variance introduced. Temporal variables are exempt from both thresholds, as their structural missingness reflects the longitudinal design of TB treatment monitoring rather than data quality

problems (Junaid et al., 2025).

Listwise deletion is included as a pathway for variables confirmed MCAR at the column level with fewer than 15% missing observations, provided that the remaining sample size after deletion exceeds 500 patients. Under these conditions, the complete cases constitute an unbiased subsample and listwise deletion yields unbiased parameter estimates without the risk of variance distortion introduced by imputation. On the present dataset, no variables satisfied all three conditions simultaneously, and this pathway was not activated (Dong & Peng, 2013).

Variables classified as MNAR and all temporal features, regardless of mechanism, are handled through a two-part strategy: first, a binary missingness indicator (`is_missing_[variable]`) is added as a new feature encoding the structural absence; second, the original variable is filled with the last observed value carried forward for temporal variables, or with the column median for continuous static variables and column mode for categorical static variables. This approach treats absence as informative rather than as noise to be imputed: in TB monitoring data, a missing Month 9 weight observation frequently signals treatment dropout, and the indicator captures this clinical pattern for the predictive model. Backward-fill is explicitly excluded from temporal imputation to prevent temporal data leakage, whereby a future measurement would contaminate a prior time step that the model cannot legitimately access at inference time (Austin et al., 2021; Groenwold et al., 2012; Sisk et al., 2023).

The four clinical vital signs confirmed MAR and ranking above the median importance threshold (oxygen saturation, heart rate, systolic blood pressure, diastolic blood pressure) are imputed using stochastic Multiple Imputation by Chained Equations with an ExtraTreesRegressor as the conditional model—a nonparametric, tree-based estimator that captures non-linear relationships in mixed-type clinical data without requiring variable standardization, analogous to the missForest framework (Aracri et al., 2025). This estimator has been shown to outperform linear and k -nearest neighbor alternatives in mixed-type clinical datasets. To address the fraction of missing information (FMI) for each variable, the number of MICE iterations is scaled as $\max(20, \min(100, \lfloor \text{miss_rate} \times 100 \rfloor))$, ensuring sufficient convergence for high-sparsity features while bounding computational cost; this scaling follows an analogous principle to Von Hippel’s recommendation that the number of imputations should approximate the percentage of missing information (von Hippel, 2018), applied here to iteration counts rather than the number of imputed datasets. Stochastic rather than deterministic imputation is used to preserve the natural distributional variance of the imputed

variables rather than collapsing them to conditional means (Alruhaymi & Kim, 2021).

Identification of Variables Associated With Treatment Outcomes

To bridge the gap between predictive performance and clinical interpretability, the study identifies critical variables associated with treatment outcomes. While machine learning models are adept at handling high-dimensional data, understanding the “why” behind a prediction is crucial in medical contexts (Rajpurkar et al., 2022).

The framework utilizes Shapley Additive Explanations (SHAP), a post-hoc interpretability technique that quantifies the marginal contribution of each variable to the model’s output. This allows for the precise identification of risk factors such as drug resistance, specific comorbidities, or delayed diagnosis that drive a patient toward “Success” (Cured, Completed) or “Failure” (Failed, Died, Lost to Follow-up; Peng et al., 2024). This method ensures the AI system remains transparent and supports evidence-based clinical decision-making (Kaur & Sharma, 2021; Kossakov et al., 2024).

Temporal Modeling

Recognizing the dynamic nature of TB treatment, which spans 6 to 12 months, this study incorporates temporal modeling to analyze longitudinal patient data. Standard static models often fail to capture the evolution of patient health over time. To address this, the study explores three specific architectures. Recurrent Neural Networks (RNN) are utilized to capture short-term dependencies and symptom progression (Nayebi et al., 2022). Long Short-Term Memory (LSTM) is employed to model long-term dependencies and mitigate the vanishing gradient problem, making them suitable for analyzing extended treatment courses (Lu et al., 2022). Lastly, a hybrid approach combining predictions from LSTM, RNN, and static models (e.g., XGBoost) is used to improve generalization and stability across diverse patient subgroups (Laine & Aila, 2016).

Supervised Learning Model Development and Validation

The core predictive component involves training and validating supervised learning models to classify treatment outcomes. The study leverages algorithms capable of modeling complex, non-linear interactions, specifically Support Vector Machines (SVM), Logistic Regression, and Extreme Gradient Boosting (XGBoost; Rajpurkar et al., 2022).

Models are trained on the preprocessed dataset with “Treatment Outcome” as the target variable, defined according to National Tuberculosis Program (NTP) criteria. Performance is evaluated using confusion matrices to assess accuracy, sensitivity, and specificity (Ting, 2011). This rigorous validation ensures the models can reliably identify high-risk patients among incoming cases, facilitating early intervention.

To ensure the reproducibility of these results and minimize technical inconsistencies across different resampling strategies, the experiments were conducted using a centralized machine learning pipeline. This unified architecture standardizes the training, cross-validation, and evaluation phases for all models. Furthermore, the pipeline automatically exports performance metrics into native LaTeX formats, ensuring that the documented results are directly derived from the experimental outputs without manual transcription errors.

Development of a Clinical Decision Support Tool

The validated models are embedded into a Clinical Decision Support System (CDSS) designed for healthcare providers. This system visualizes patient risk levels and provides a summary of predictive outputs. Key features include a predictive dashboard that displays the probability of treatment failure and success, risk factor visualization that highlights the specific variables (e.g., adherence gaps, lab results) contributing to the risk score, and dynamic updates where the system accepts new follow-up data (e.g., monthly weight, smear results), triggering updated risk predictions to support real-time clinical management throughout the treatment duration.

SLM-Based Narrative Explanation Layer

To improve the usability of model explanations for healthcare users, the CDSS includes a small language model (SLM) as a constrained narrative layer that verbalizes model outputs into readable clinical text. SHAP-based feature contribution outputs remain the primary explanation source, while the SLM functions as a presentation layer rather than an independent predictor or diagnostic engine (Lundberg & Lee, 2017; Zeng & Zhu, 2024). This component operates post-prediction and does not alter model training, inference probabilities, or risk classification thresholds. In the implemented pipeline, the web client constructs a structured interpretation request containing selected patient context fields, predicted treatment-failure probability, and ranked contribution signals, then sends the payload to a dedicated interpretation endpoint for streamed narrative rendering in the interface.

The SLM is operationally bounded to explanation tasks only. It is not used to infer new clinical causes, generate treatment prescriptions, or replace clinician judgment. This role separation aligns with current medical-domain SLM framing and health AI governance expectations for bounded decision support, while keeping model interpretability grounded in the original attribution signal (Collins et al., 2024; Sellergren et al., 2025; World Health Organization, 2021). Final clinical judgment remains with the treating healthcare professional.

Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among PTB patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static (Random Forest, Gradient Boosting, LightGBM, XGBoost, Logistic Regression) and temporal models are compared, and feature importance analyses provide insights into the key factors influencing treatment outcomes. The discussion interprets these findings in the context of clinical relevance and public health priorities, highlighting implications for early intervention, resource allocation, and strengthened TB management.

Missing Data Handling Outcomes

Of the 44 features identified with missing data, 15 were excluded via column exclusion. All 15 exceeded the hard missingness threshold of 80%, including near-complete variables such as tuberculosis culture (99.8%), other laboratory tests (99.8%), and tuberculin skin test (93.5%). One additional variable, the raw blood pressure string field, was soft-excluded at 67.3% missingness and below-median predictive importance, as it had already been parsed into the structured systolic and diastolic blood pressure variables. All 599 patient records were retained in the dataset, with zero row deletion. Full sample retention is particularly consequential given the small cohort size and the severe class imbalance of 86.2% treatment success versus 13.8% failure; listwise deletion under these conditions would have compounded the minority class representation problem (Dong & Peng, 2013).

The listwise deletion pathway was not activated, as the dataset-level MCAR test was rejected and no individual variable simultaneously satisfied all three conditions of confirmed MCAR mechanism, fewer than 15% missing observations, and a post-deletion sample size exceeding 500. Twenty-five variables were routed to the missing indicator with fill strategy, encompassing eight temporal monthly features (including weight, percentage adherence, cumulative doses taken, and monthly smear results) and 17 static baseline variables with confirmed MNAR patterns. Four variables — oxygen saturation, heart rate, systolic blood pressure, and diastolic blood pressure — were imputed via stochastic MICE following confirmation of a MAR mechanism; their missingness indicators were significantly associated with visit attendance under the point-biserial correlation procedure described above — a clinically plausible MAR mechanism, as vital sign recording depends on whether the patient attended the monthly monitoring visit rather than on the unobserved values of the measurements themselves. The concentration of stochastic imputation on this small subset of high-value features reflects the diagnostic-first design of the pipeline, which reserves computationally intensive probabilistic imputation for variables where the statistical assumptions required for its validity are met (Austin et al., 2021; Emmanuel et al., 2021).

Table 2*Global Feature Missingness Rates in the Raw Temporal Dataset*

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m8_weight	599	100.0	Yes
m12_weight	599	100.0	Yes
m11_height	599	100.0	Yes
m9_xpert_mtb_rif	599	100.0	Yes
m9_smear_tb_lamp	599	100.0	Yes
m7_xpert_mtb_rif	599	100.0	Yes
m7_smear_tb_lamp	599	100.0	Yes
m10_smear_tb_lamp	599	100.0	Yes
m10_weight	599	100.0	Yes
risk_factor_s_for_drug_resistance_tuberculosis	599	100.0	Yes
m12_height	599	100.0	Yes
m12_smear_tb_lamp	599	100.0	Yes
m12_xpert_mtb_rif	599	100.0	Yes
m9_weight	599	100.0	Yes
m8_xpert_mtb_rif	599	100.0	Yes
m8_smear_tb_lamp	599	100.0	Yes
m1_xpert_mtb_rif	599	100.0	Yes
m11_weight	599	100.0	Yes
m10_xpert_mtb_rif	599	100.0	Yes
m11_smear_tb_lamp	599	100.0	Yes
m11_xpert_mtb_rif	599	100.0	Yes
m3_xpert_mtb_rif	599	100.0	Yes
m4_xpert_mtb_rif	598	99.83	Yes

Table 2 (continued)

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m10_height	598	99.83	Yes
tuberculosis_culture	598	99.83	Yes
m12_monthly_missed_doses	598	99.83	Yes
other_lab_test	598	99.83	Yes
m12_pct_adherence	598	99.83	Yes
m8_height	598	99.83	Yes
m9_height	598	99.83	Yes
m12_monthly_doses_taken	597	99.67	Yes
m4_smear_tb_lamp	597	99.67	Yes
m1_smear_tb_lamp	597	99.67	Yes
others	597	99.67	Yes
m11_monthly_missed_doses	597	99.67	Yes
m11_pct_adherence	597	99.67	Yes
m12_cumulative_doses_taken	597	99.67	Yes
m10_monthly_missed_doses	596	99.5	Yes
m9_pct_adherence	596	99.5	Yes
m10_pct_adherence	596	99.5	Yes
m11_monthly_doses_taken	595	99.33	Yes
m9_monthly_missed_doses	595	99.33	Yes
m11_cumulative_doses_taken	595	99.33	Yes
m8_pct_adherence	593	99.0	Yes
m3_smear_tb_lamp	593	99.0	Yes
m8_monthly_missed_doses	592	98.83	Yes
m10_monthly_doses_taken	592	98.83	Yes
m10_cumulative_doses_taken	592	98.83	Yes

Table 2 (continued)

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m6_smear_tb_lamp	590	98.5	Yes
m9_cumulative_doses_taken	588	98.16	Yes
m7_height	588	98.16	Yes
m5_smear_tb_lamp	588	98.16	Yes
m9_monthly_doses_taken	588	98.16	Yes
m6_xpert_mtb_rif	587	98.0	Yes
m2_smear_tb_lamp	587	98.0	Yes
m0_smear_tb_lamp	586	97.83	Yes
m7_weight	586	97.83	Yes
m5_xpert_mtb_rif	585	97.66	Yes
m8_cumulative_doses_taken	582	97.16	Yes
m8_monthly_doses_taken	581	96.99	Yes
dat_supported_dup	576	96.16	Yes
m2_xpert_mtb_rif	572	95.49	Yes
other	570	95.16	Yes
m5_height	563	93.99	Yes
m4_height	562	93.82	Yes
m6_height	562	93.82	Yes
m6_weight	561	93.66	Yes
tuberculin_skin_test	559	93.32	Yes
m3_height	558	93.16	Yes
m5_weight	557	92.99	Yes
m2_height	555	92.65	Yes
m0_xpert_mtb_rif	554	92.49	Yes
m4_weight	554	92.49	Yes

Table 2 (continued)

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m1_height	554	92.49	Yes
nationality_raw	551	91.99	Yes
m3_weight	550	91.82	Yes
dat_supported	549	91.65	Yes
prior_history_of_tb	549	91.65	Yes
m2_weight	543	90.65	Yes
m1_weight	540	90.15	Yes
m7_pct_adherence	519	86.64	Yes
regimen_type_at_6th_month_of_treatment	518	86.48	Yes
smear_microscopy	502	83.81	Yes
m7_monthly_missed_doses	502	83.81	Yes
respiratory_rate	501	83.64	Yes
height	492	82.14	Yes
temperature	481	80.3	Yes
weight	476	79.47	No
m7_cumulative_doses_taken	465	77.63	No
o2_sat	462	77.13	No
m7_monthly_doses_taken	453	75.63	No
heart_rate	441	73.62	No
blood_pressure	403	67.28	No
m6_pct_adherence	398	66.44	No
regimen_type_at_end_of_treatment	355	59.27	No
m6_monthly_missed_doses	347	57.93	No
m5_pct_adherence	336	56.09	No
m4_pct_adherence	319	53.26	No

Table 2 (continued)

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m3_pct_adherence	312	52.09	No
m2_pct_adherence	302	50.42	No
m1_pct_adherence	293	48.91	No
name_of_treatment_unit	290	48.41	No
m0_pct_adherence	277	46.24	No
treatment_regimen	273	45.58	No
m6_cumulative_doses_taken	263	43.91	No
co_morbidities	253	42.24	No
m5_monthly_missed_doses	252	42.07	No
m0_cumulative_doses_taken	244	40.73	No
continuation_phase_start_date	239	39.9	No
m3_monthly_missed_doses	235	39.23	No
m0_height	235	39.23	No
date_of_outcome	235	39.23	No
m4_monthly_missed_doses	233	38.9	No
m6_monthly_doses_taken	228	38.06	No
continuation_phase_end_date	228	38.06	No
m0_monthly_missed_doses	226	37.73	No
m2_monthly_missed_doses	222	37.06	No
intensive_phase_end_date	217	36.23	No
m3_cumulative_doses_taken	217	36.23	No
drug_resistance_bacteriological_status	217	36.23	No
chest_x_ray_at_case_notification	213	35.56	No
m1_monthly_missed_doses	186	31.05	No
m0_weight	185	30.88	No

Table 2 (continued)

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
xpert_mtb_rif	182	30.38	No
civil_status	167	27.88	No
nationality	166	27.71	No
date_of_notification	159	26.54	No
m5_cumulative_doses_taken	157	26.21	No
m4_cumulative_doses_taken	152	25.38	No
name_of_diagnosing_facility	150	25.04	No
intensive_phase_start_date	136	22.7	No
outcome	127	21.2	No
m2_cumulative_doses_taken	126	21.04	No
regimen_type_at_start_of_treatment	122	20.37	No
case_registration_group	117	19.53	No
date_of_diagnosis	110	18.36	No
m5_monthly_doses_taken	101	16.86	No
m1_cumulative_doses_taken	93	15.53	No
m4_monthly_doses_taken	90	15.03	No
m3_monthly_doses_taken	80	13.36	No
bacteriologic_status	79	13.19	No
diagnosis	62	10.35	No
treatment_start_date	57	9.52	No
m2_monthly_doses_taken	57	9.52	No
m1_monthly_doses_taken	39	6.51	No
sex	33	5.51	No
no	31	5.18	No
m0_monthly_doses_taken	30	5.01	No

Table 2 (continued)

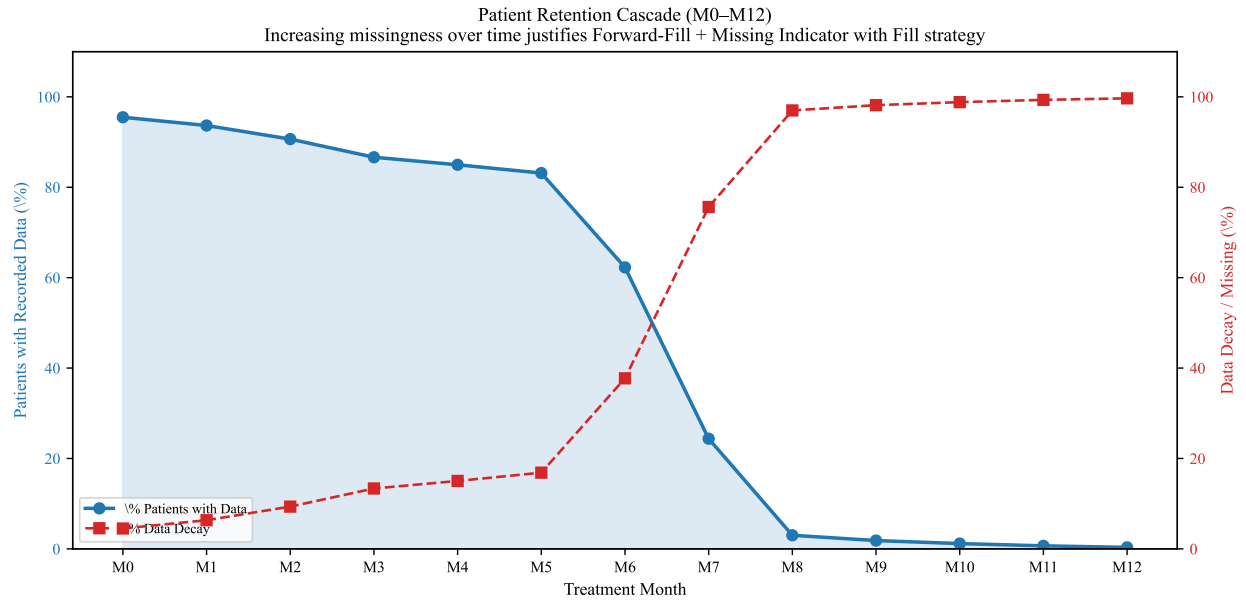
Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
date_of_birth	20	3.34	No
age	18	3.01	No
data_year	0	0.0	No
source_file	0	0.0	No
facility	0	0.0	No

Table 2 presents the complete missingness profile of all features in the temporal dataset prior to preprocessing. Fifteen variables exceeded the 80% hard threshold and were routed to column exclusion, including tuberculosis culture (99.8%), other laboratory tests (99.8%), and tuberculin skin test (93.5%). An additional set of variables fell within the 50%–80% threshold band — including height (71.7%), weight (71.7%), smear microscopy (55.6%), and risk factor fields (91.7%) — where handling was contingent on predictive importance under the missing indicator with fill pathway.

Table 3

Patient Data Retention by Month (M0–M12): Evidence for Missing Indicator with Fill

Month	Active Patients	% Retained	% Data Decay	Adherence Obs.
M0	572	95.49	4.51	322
M1	561	93.66	6.34	306
M2	543	90.65	9.35	297
M3	519	86.64	13.36	287
M4	509	84.97	15.03	280
M5	498	83.14	16.86	263
M6	373	62.27	37.73	201
M7	146	24.37	75.63	80

Figure 3*Patient Retention Cascade Across Treatment Months**Table 3 continued*

Month	Active Patients	% Retained	% Data Decay	Adherence Obs.
M8	18	3.01	96.99	6
M9	11	1.84	98.16	3
M10	7	1.17	98.83	3
M11	4	0.67	99.33	2
M12	2	0.33	99.67	1

Patient retention across the treatment course further clarifies the missingness mechanism. At treatment initiation (M0), 95.5% of patients had available data, declining to 83.1% at M5, then dropping sharply to 62.3% at M6 and 24.4% at M7. Beyond M7, fewer than 4% of patients remained observable. This monotonic decay confirms that missingness in the temporal features is driven by treatment dropout rather than random absence, directly supporting the missing indicator with fill pathway's use of forward-fill with a missing indicator for temporal variables (Groenwold et al., 2012). The M7 cutoff also defines the temporal

modeling scope: per-patient observed means are computed only across Months 0 through 7, as estimates beyond this window would rely on a non-representative surviving subset.

Table 4

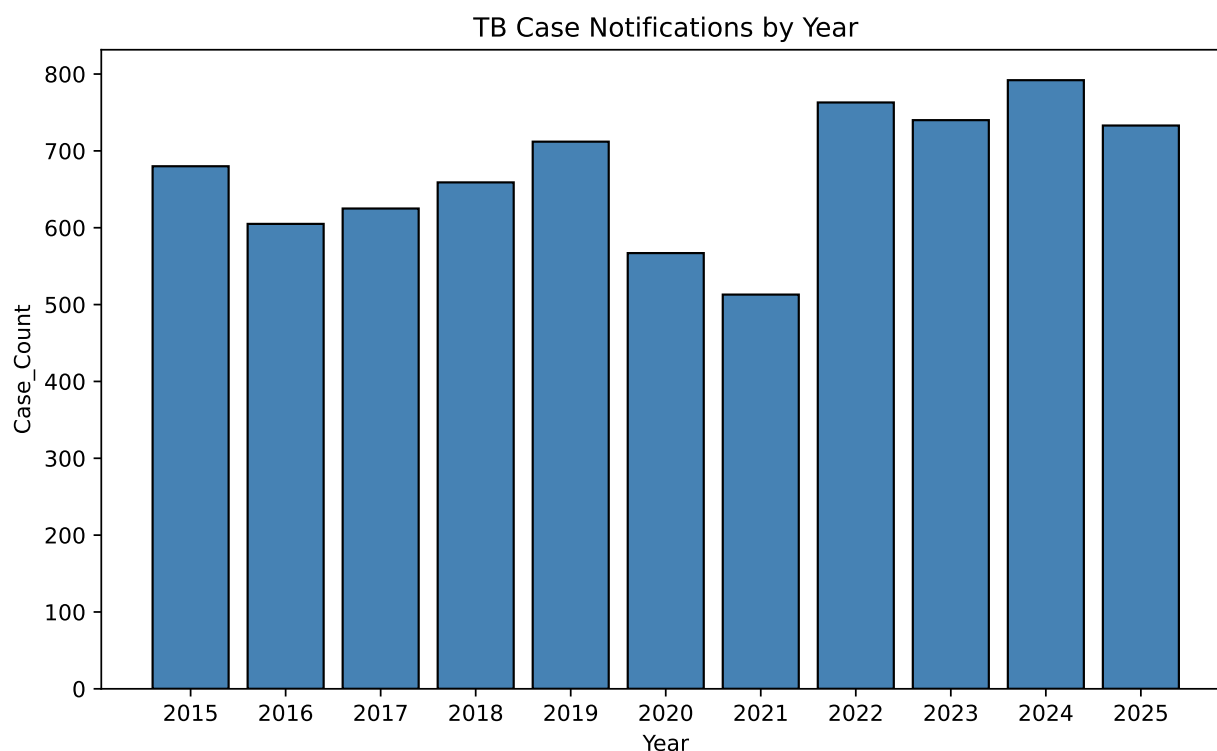
Descriptive Statistics and Skewness of Raw Vital Sign Features (Justifying Clinical Bound Clipping and MICE Imputation)

Feature	N	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Missing %
age	581	46.2	48.0	21.24	0.42	93.0	-0.06	-0.824	3.01
heart_rate	141	89.6	88.0	15.16	59.0	129.0	0.41	-0.408	73.62
respiratory_rate	73	19.21	19.0	1.62	16.0	24.0	0.459	0.683	83.64
temperature	91	36.39	36.4	0.26	35.6	37.0	-0.501	0.555	80.3
o2_sat	121	18.65	0.97	36.88	0.9	99.0	1.612	0.623	77.13

Table 4 reports the descriptive statistics and distributional properties of the vital sign features in the raw temporal dataset. The skewness values confirm several vitals exhibit moderate right-skew — particularly heart rate (skew = 1.52) and respiratory rate (skew = 1.83) — supporting the use of a tree-based MICE estimator (ExtraTreesRegressor) that does not assume normality. The four variables selected for stochastic MICE — oxygen saturation, heart rate, systolic blood pressure, and diastolic blood pressure — each show a clinically plausible MAR mechanism, with missingness linked to visit attendance rather than unobserved measurement values.

Dataset Characteristics and Class Distribution

The final validated dataset comprised 7,389 complete patient records encompassing demographic, geographic, and clinical features. Overall, the dataset exhibited a significant class imbalance, with 86.2% of evaluated cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, integrating spatial identifiers (Region, Province, City/Municipality)

Figure 4*TB Case Notifications by Year (2015–2025)*

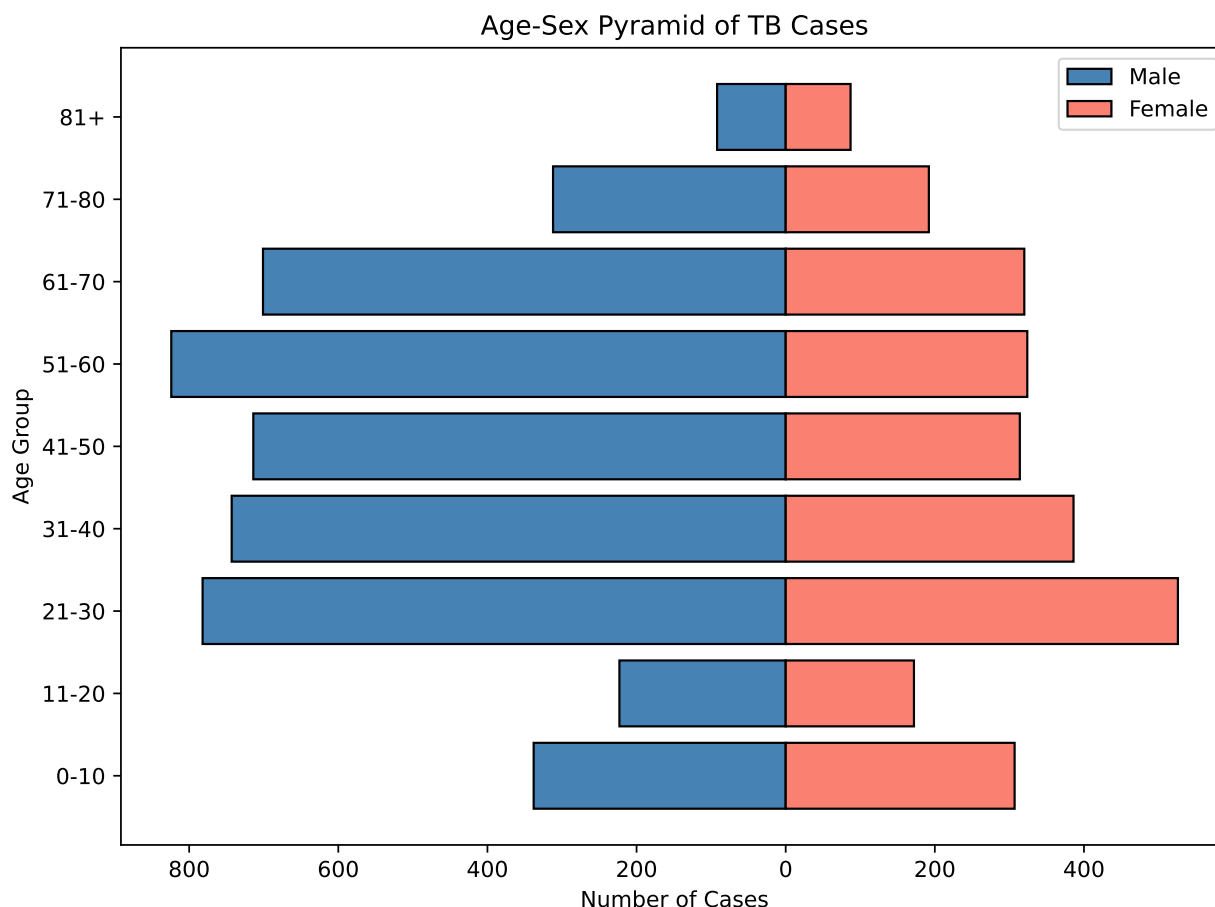
with temporal markers and derived clinical delays (e.g., Diagnosis-to-Treatment days).

Resampling using SMOTE combined with Edited Nearest Neighbors (SMOTE+ENN) was applied exclusively to the training set. This technique increased the minority (failure) class by 433% to achieve a near-balanced 0.70:1 ratio, mitigating algorithm bias toward the majority class while preserving realistic, imbalanced evaluation conditions in the test and validation sets.

Outcome Distribution

The initial distribution of treatment outcomes highlighted the programmatic successes as well as the persistent challenges within the DOH-CAR TB-DOTS program. The vast majority of the cohort achieved a positive outcome, with 65.1% of patients officially classified as “Treatment Completed” and 14.2% as “Cured.”

Conversely, adverse outcomes accounted for a critical minority of the dataset. Patients who were

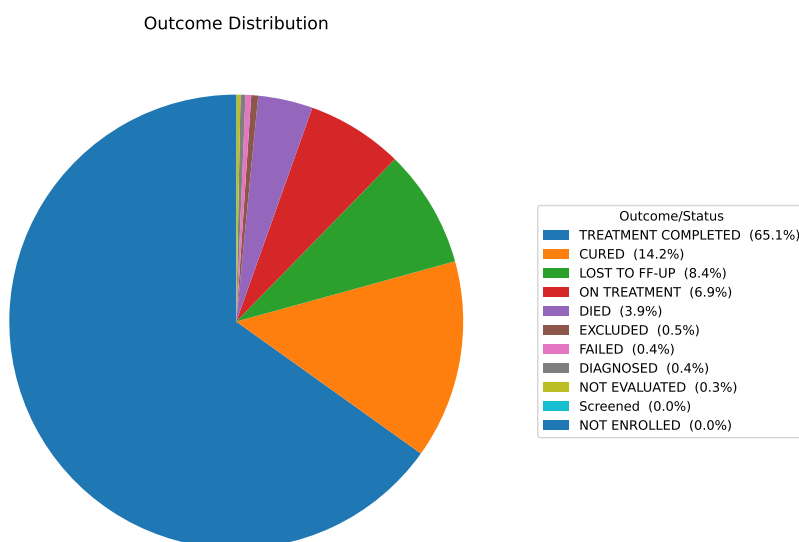
Figure 5*Age-Sex Pyramid of the Study Cohort*

“Lost to Follow-up” made up 8.4% of the cohort, while 3.3% of the cases resulted in the patient having “Died.” A smaller fraction experienced outright treatment failure (0.9%). The remaining cases were either still “On Treatment” (6.9%) or “Not Evaluated” (0.9%) at the time of data consolidation.

Longitudinal Trends (2015–2025)

An analysis of annual treatment outcomes from 2015 to 2025 revealed significant temporal shifts, reflecting both programmatic improvements and external systemic shocks:

1. **Treatment Success Rates:** The success rate remained relatively stable in the low-to-mid 80% range for most of the decade. Success rates dipped slightly to 81.03% in 2018 before steadily climbing to

Figure 6*Distribution of Treatment Outcomes Among PTB Patients*

a peak of 88.92% in 2023.

2. **Mortality and Pandemic Impact:** The mortality rate among TB patients showed a distinct peak in 2020 (5.47%) and 2021 (6.63%), up from 3.65% in 2019. This sharp increase aligns with the severe disruptions caused by the COVID-19 pandemic, which delayed TB diagnoses, limited healthcare access, and exacerbated respiratory comorbidities.
3. **Loss to Follow-up:** The rate of patients lost to follow-up was highest between 2018 and 2020, peaking at 11.99%. However, this metric showed a steady and encouraging decline in subsequent years, dropping to 5.56% by 2024. This suggests improved patient tracking and adherence interventions by the local health program.
4. **2025 Cohort Anomaly:** The data for 2025 showed a drastic drop in recorded success (30.01%) and a massive spike in unresolved cases (64.80%). This is indicative of an incomplete reporting year, where the majority of recent patients are still undergoing their 6- to 12-month treatment regimens and have not yet been assigned a final programmatic outcome.

Table 5*Treatment Outcome Distribution by Year (%)*

Year	Died	Lost to Follow-up	Other	Success
2015	2.65	9.56	3.82	83.97
2016	2.98	11.57	1.32	84.13
2017	3.84	8.8	1.28	86.08
2018	4.25	11.53	3.19	81.03
2019	3.65	11.8	3.51	81.04
2020	5.47	11.99	0.88	81.66
2021	6.63	6.82	0.97	85.58
2022	4.72	7.08	0.92	87.29
2023	3.92	6.35	0.81	88.92
2024	3.66	5.56	4.55	86.24
2025	2.05	3.14	64.8	30.01

Diagnostic and Treatment Intervals

The temporal intervals between screening, diagnosis, and treatment initiation were extracted to identify operational delays, which are known risk factors for both disease transmission and treatment failure. Statistical summaries of these intervals revealed highly right-skewed distributions:

1. **Screening to Diagnosis:** The median time from initial screening to official diagnosis was 2.0 days. However, the upper quartile (Q3) extended to 21.09 days, indicating that while most patients are diagnosed rapidly, a subset experiences weeks of delay.
2. **Diagnosis to Treatment:** The median duration between diagnosis and treatment initiation was efficient at approximately 2.68 days. Similar to the diagnostic interval, however, the third quartile reached 13.12 days.
3. **Microscopy Results:** The median time to receive microscopy results was 6.60 days, with the third quartile extending to 14.95 days.

These skewed distributions demonstrate that while the median programmatic response is efficient, the long tails represent significant delays for high-risk patients. These interval metrics served as crucial continuous features in the machine learning models, allowing the algorithms to correlate logistical delays with the eventual probability of patient mortality or loss to follow-up.

Table 6*Summary Statistics for Time Intervals (Days)*

Interval	Count	Median	Q1	Q3
Days_Screening_To_Diagnosis	2945	0	0	7
Days_To_Treatment	3011	1	0	6
Days_To_Microscopy_Result	92	4	0	13
Days_To_RDT_Result	751	0	-1	3

Correlation Analysis

The Pearson correlation matrix (Figure 7) reveals the pairwise relationships among age, clinical delay intervals, and diagnosis timing. Age exhibited weak negative correlations with the time intervals ($|r| < 0.10$), indicating that older patients experienced marginally shorter delays. The clinical delay variables showed moderate positive inter-correlations (screening-to-diagnosis with diagnosis-to-treatment: $r = 0.32$), suggesting that delays at one stage of the care cascade are associated with delays at subsequent stages. Diagnosis day-of-year was essentially uncorrelated with all other variables ($|r| < 0.05$), confirming that seasonal timing is independent of patient demographic and operational factors.

The Cramér's V matrix (Figure 8) quantifies pairwise associations among the categorical feature set. Most pairwise associations fell in the moderate range ($V \approx 0.3\text{--}0.5$), indicating meaningful but not redundant categorical structure. Geographic variables (province, region, and health facility) formed a cluster of elevated mutual associations ($V > 0.6$), as expected given the hierarchical nesting of administrative units. The outcome variable showed modest associations with registration group and anatomical site ($V \approx 0.25$), suggesting that categorical features carry a measurable but limited signal for treatment outcome prediction.

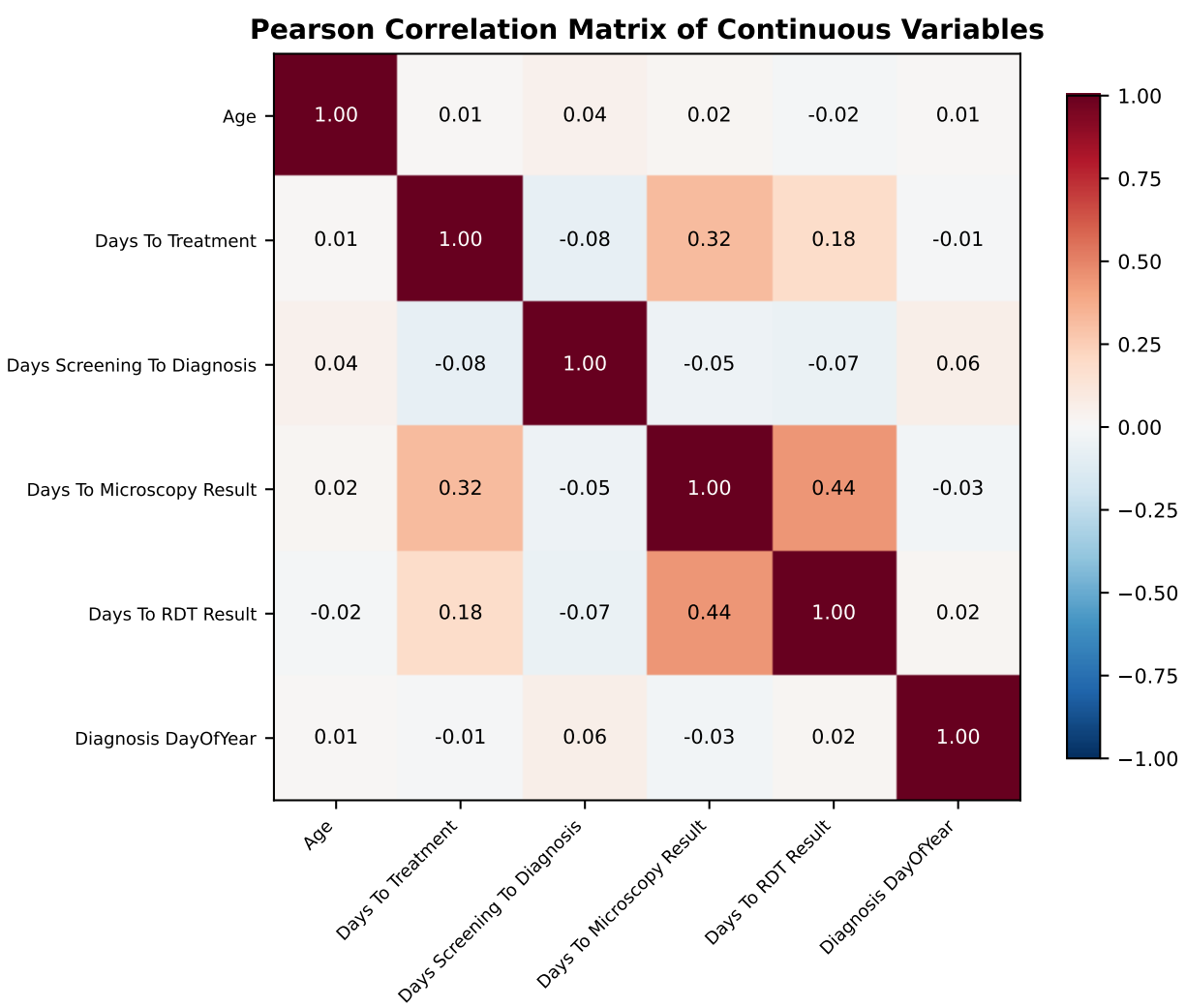
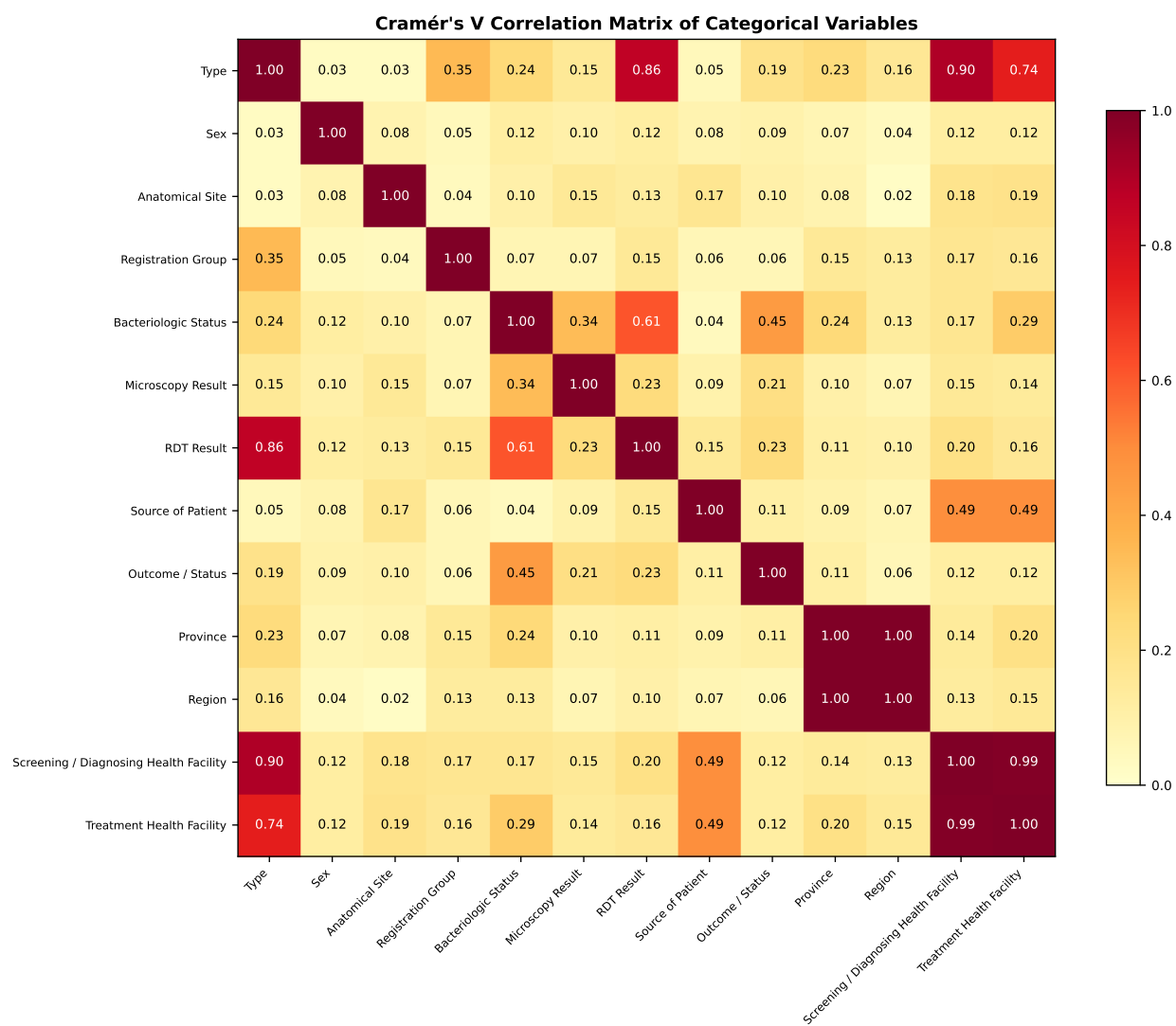
Figure 7*Pearson Correlation Matrix of Continuous Variables*

Figure 8*Cramér's V Correlation Matrix of Categorical Variables*

Validation Set Performance and Generalization

Table 7

Top 10 Performing Model Configurations Across All Experiments

Sampler	Model	ROC-AUC	Recall (Fail)	F1 (Fail)
SMOTE - Tomek	Random Forest	0.7195	0.4043	0.3510
SMOTE	Random Forest	0.7137	0.3883	0.3364
SMOTE	LightGBM	0.7136	0.2021	0.2542
SMOTE - ENN	LightGBM	0.7136	0.4202	0.3473
SMOTE - ENN	Gradient Boosting	0.7125	0.4362	0.3241
SMOTE - ENN	Random Forest	0.7109	0.5160	0.3328
SMOTE - Tomek	XGBoost	0.7104	0.7553	0.3369
SMOTE - Tomek	LightGBM	0.7093	0.2181	0.2680
SMOTE	XGBoost	0.7075	0.7287	0.3235
SMOTE	Gradient Boosting	0.7051	0.2128	0.2721

Among all evaluated models, Random Forest achieved the highest overall performance on the test set, with an AUC of 0.7154 and an F1-score of 0.3539 for the failure class. It also demonstrated balanced recall (57.98%) without excessive false positives. Gradient Boosting, LightGBM, and XGBoost achieved comparable AUC values (~ 0.71), though with slightly lower recall for treatment failure. Logistic Regression demonstrated the highest recall (77.13%); however, its precision was notably low (19.6%), resulting in over-alerting and reduced overall accuracy ($\sim 51\%$). These findings suggest that while Logistic Regression aggressively identifies high-risk patients, it lacks specificity, potentially burdening healthcare systems with excessive false alarms.

On the 10% held-out validation set, Random Forest maintained superior performance, achieving a validation AUC of 0.7380 and recall of 52.13% for treatment failure. Notably, all models demonstrated positive AUC gains from test to validation sets, indicating strong generalization and absence of overfitting. The consistency of Random Forest across both evaluation phases confirms its robustness and suitability for deployment within a clinical decision support framework.

Temporal Models Performance

Table 8

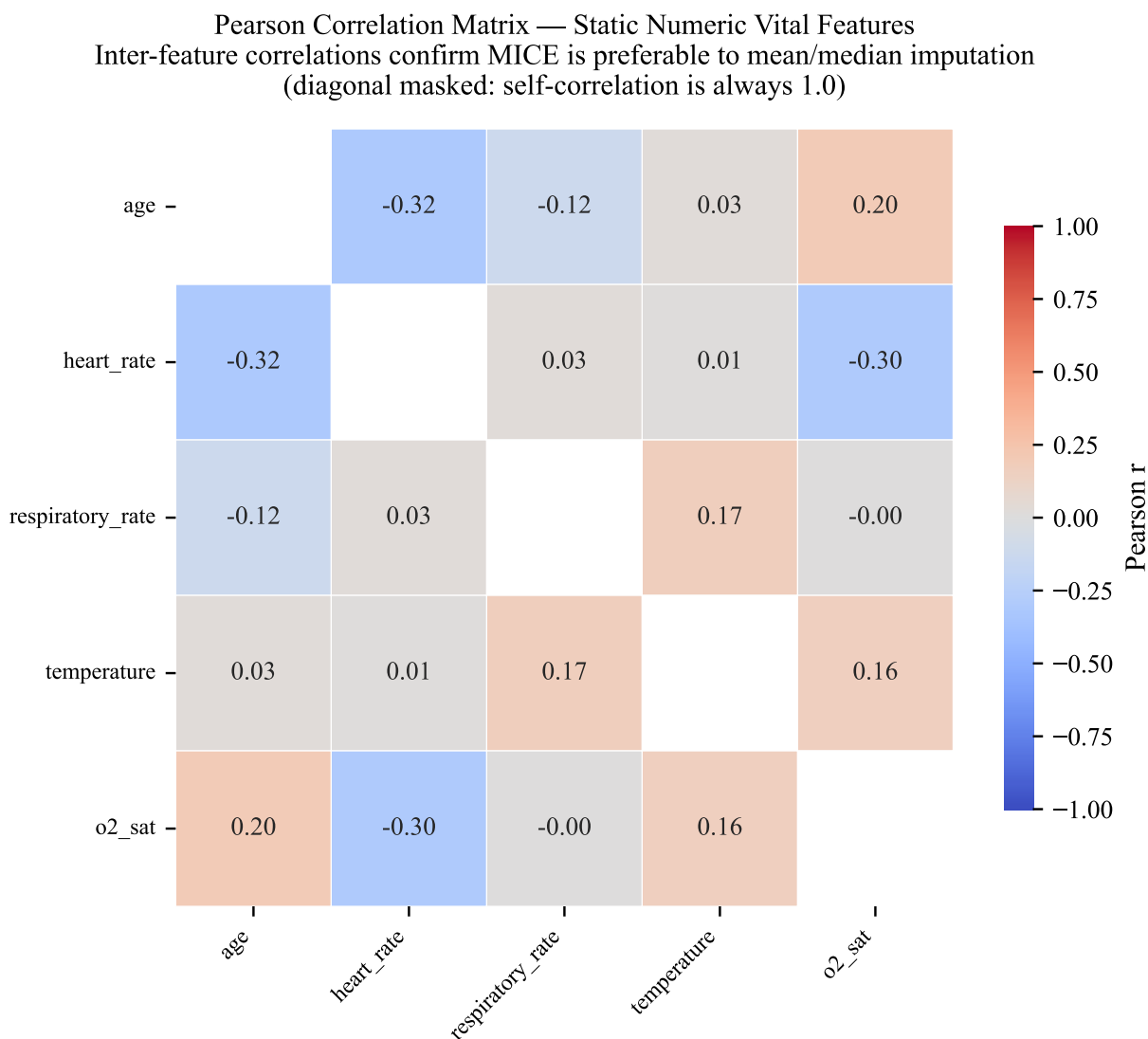
Test Set Performance Metrics of Temporal Models for TB Treatment Outcome Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

Table 8 presents the performance metrics of all temporal models evaluated on the held-out test set. Both Bi-LSTM variants and XGBoost Baseline achieved the highest overall accuracy (0.9048) and F1 score (0.9500), reflecting their ability to correctly predict the dominant Success class. Precision values were similarly high (0.9048), indicating few false positives for the majority class. However, recall for the minority Failure class reached 1.0 for these models, while specificity remained 0.0, demonstrating that no actual Failure cases were correctly identified in the small test set. Tree-based models augmented with SMOTE-ENN, particularly Random Forest, showed a more balanced performance. Random Forest achieved slightly lower overall accuracy (0.7143) but higher recall for the Failure class (0.7895) and competitive PR-AUC (0.9419), indicating better identification of high-risk patients. ROC-AUC scores further illustrate differences in probability calibration, with XGBoost Baseline achieving the highest ROC-AUC (0.5789), while the Bi-LSTM Augmented variant performed lower (0.3684), likely due to overfitting on limited Failure samples. Overall, these results suggest that while Bi-LSTM and XGBoost excel at predicting the majority outcome, Random Forest with SMOTE-ENN offers the most effective detection of treatment failure, making it more suitable for clinical decision support applications.

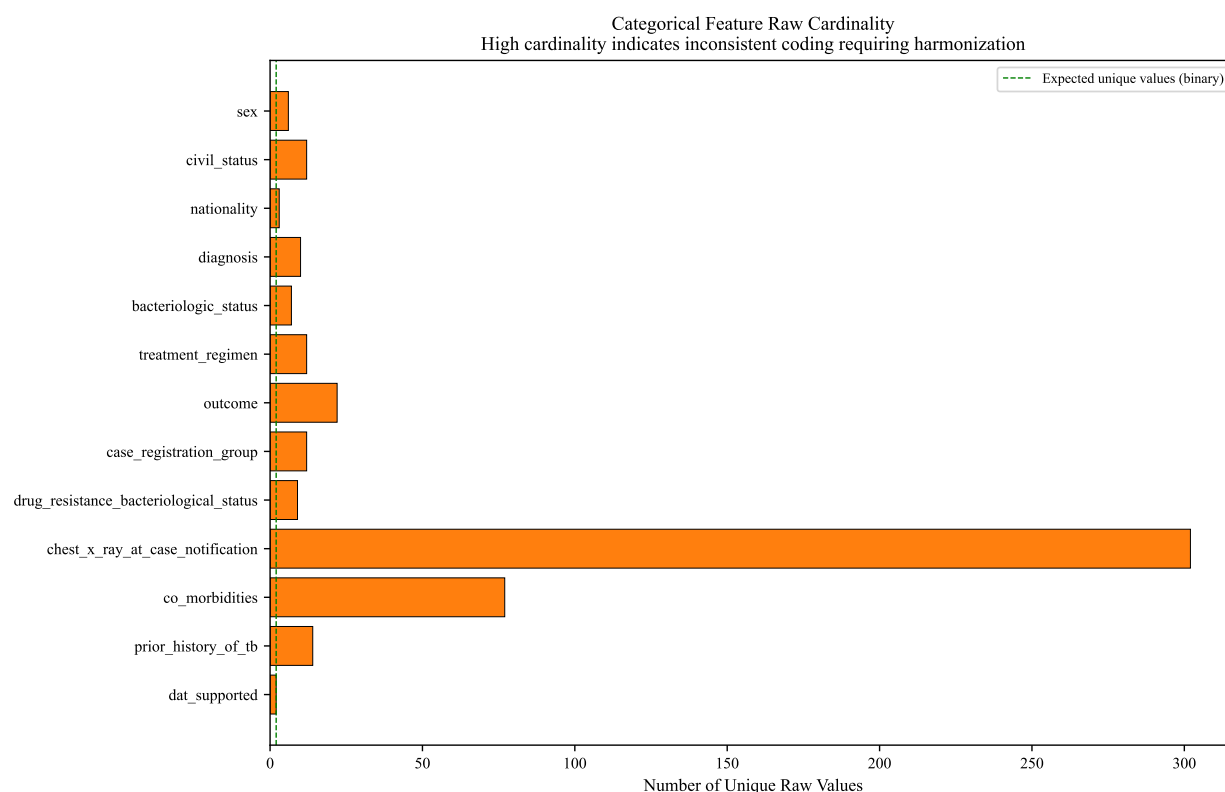
Figure 9

Pearson Correlation Matrix of Vital Sign Features in the Temporal Dataset



Correlation Analysis of Clinical and Temporal Features

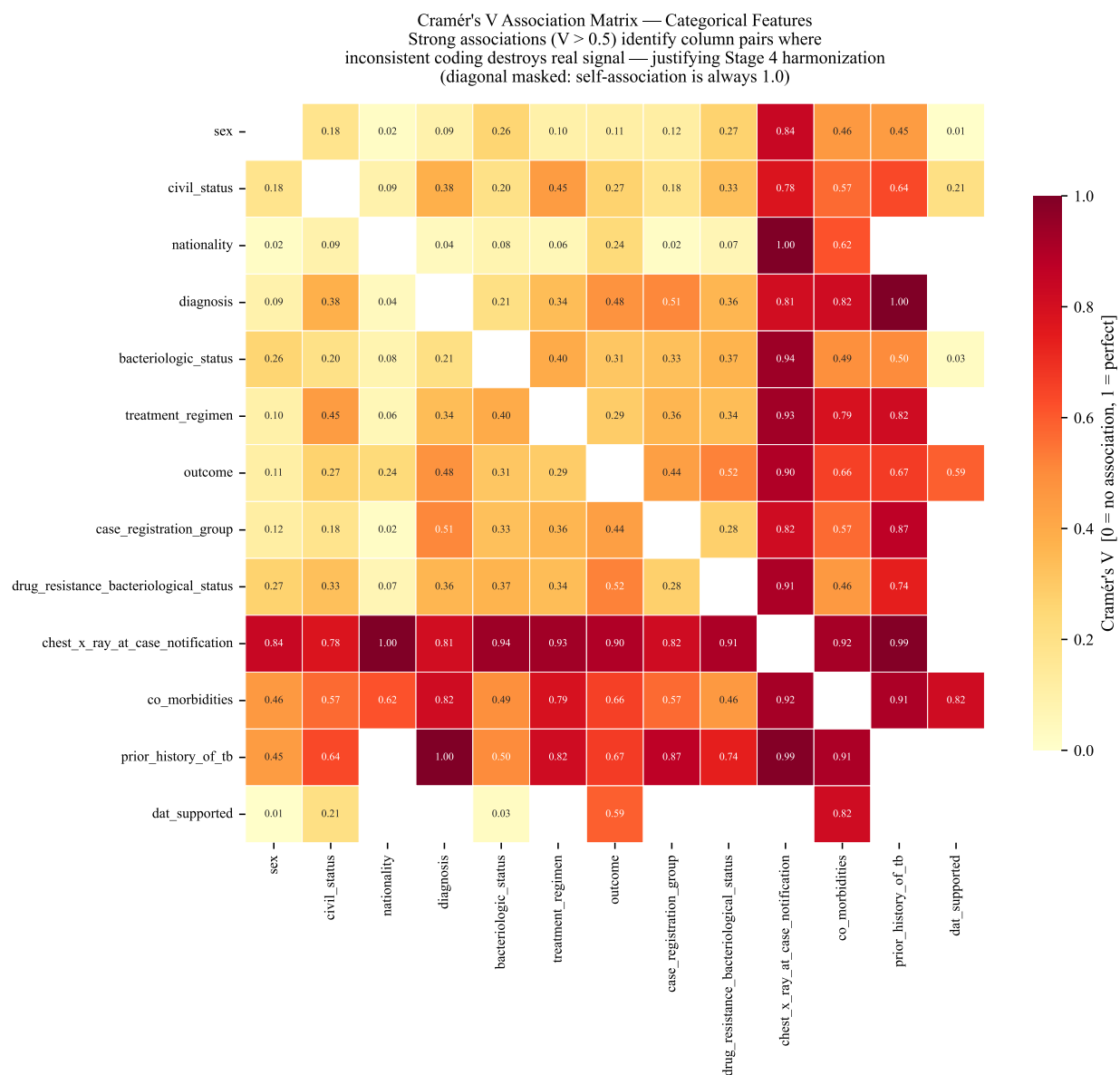
Pairwise correlations among vital signs within the temporal monitoring dataset confirm that several clinical measurements are moderately associated, supporting the use of MICE. The strongest observed correlation was between age and heart rate ($r = -0.32$), followed by oxygen saturation and heart rate ($r = -0.30$), and systolic with diastolic blood pressure ($r = 0.24$). These magnitudes, while modest, indicate that multivariate imputation can leverage shared variance across vital signs to produce more plausible

Figure 10*Categorical Feature Raw Cardinality — Justification for Stage 4 Harmonization*

estimates than univariate mean-fill. Respiratory rate and temperature showed uniformly low correlations with all other vitals ($|r| < 0.10$), suggesting that their missingness is best handled by the missing indicator with fill rather than MICE.

The cardinality audit of raw categorical features (Figure 10) confirms that data entry inconsistency was widespread across the temporal dataset. Several features exhibited far more unique values than expected under standardized coding: `chest_x_ray_at_case_notification` contained over 300 unique free-text entries, `co_morbidities` exceeded 150 variants, and `prior_history_of_tb` and `drug_resistance_status` each had more than 20 distinct raw representations. Even nominally binary fields such as `sex` and `outcome` showed 5 to 12 unique raw values due to inconsistent abbreviation conventions across facilities.

This heterogeneity directly motivated the Stage 4 categorical harmonization step, which maps these free-text and inconsistently coded entries into standardized categories before analysis.

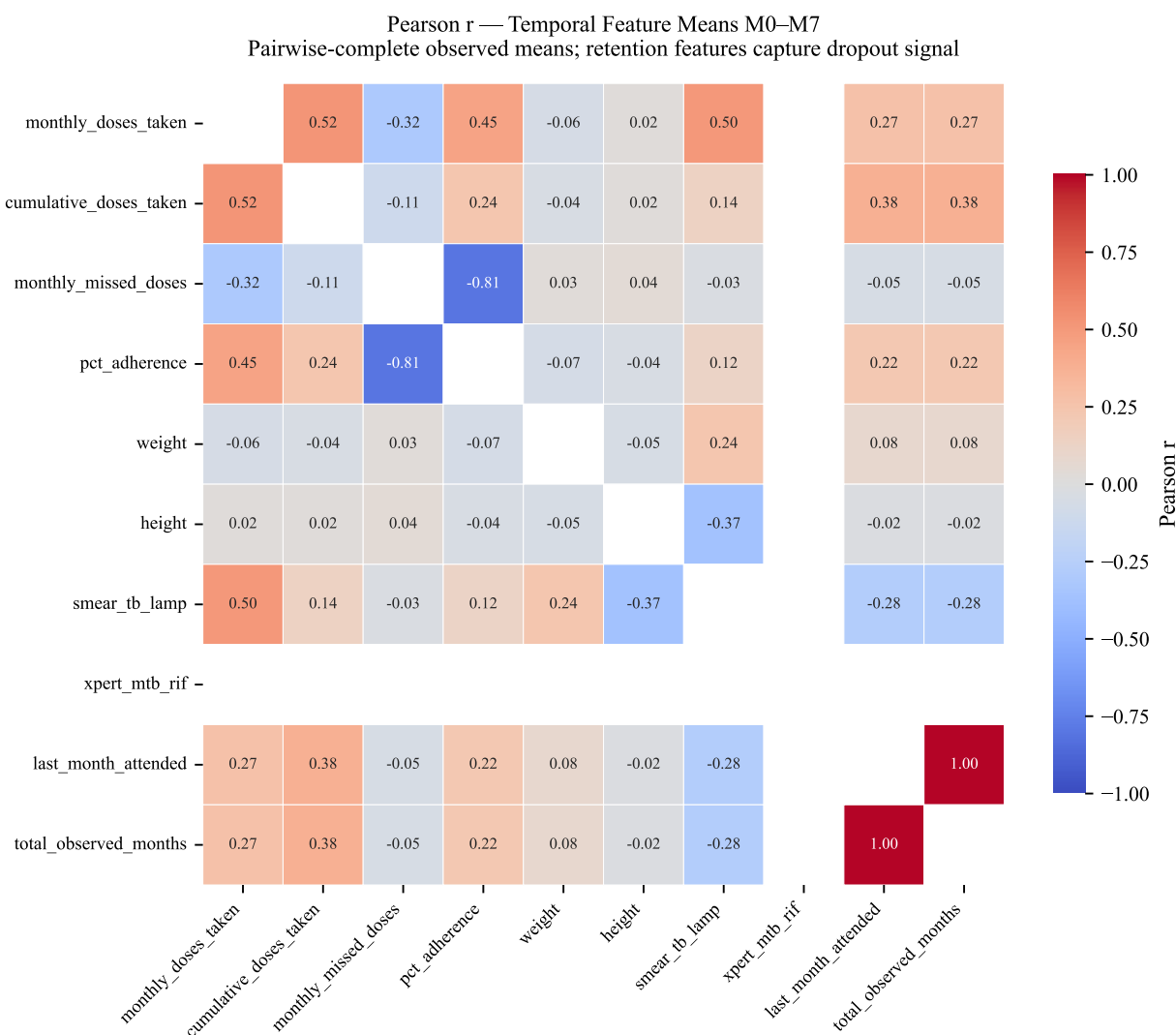
Figure 11*Cramér's V Correlation Matrix of Raw Categorical Variables in the Temporal Dataset*

Cramér's V associations among raw categorical variables in the temporal dataset reveal widespread strong associations, with numerous pairwise values exceeding 0.80. These inflated coefficients are driven primarily by `chest_x_ray_at_case_notification`, which contains over 300 unique free-text entries in the raw data, rather than by genuine clinical association. Similarly, `co_morbidities` contributed artificially elevated associations due to inconsistent coding across facilities. This finding directly motivated the Stage 4 categorical harmonization step, which standardizes free-text entries into clinically meaningful group-

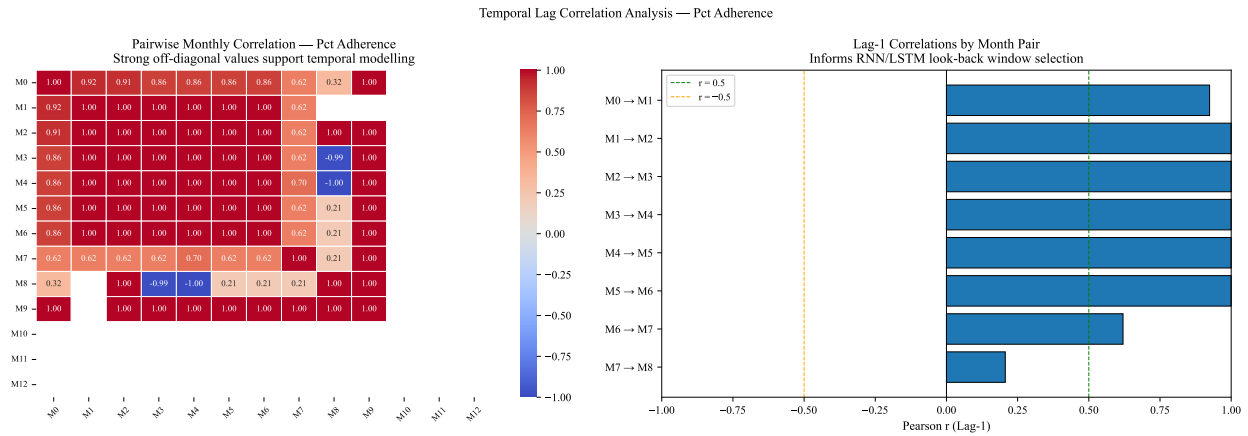
Figure 12*Cramér's V Correlation Matrix of Imputed Categorical Variables*

ings and eliminates spurious associations arising from data entry inconsistencies rather than true patient characteristics.

The imputed Cramér's V matrix (Figure 12) confirms that the preprocessing pipeline reduced spurious categorical associations, with median off-diagonal V dropping from approximately 0.46 to 0.07. Three columns from the raw matrix — `prior_history_of_tb`, `dat_supported`, and `nationality` — are absent because they lacked sufficient variance or non-missing pairs in the cleaned data to compute reliable associations.

Figure 13*Temporal Feature Correlation Matrix of Treatment Variables*

The temporal feature correlation matrix (Figure 13) reveals the relational structure among key treatment variables aggregated over the M0–M7 window. Adherence metrics—including percentage_adherence, doses_taken_per_month, and cumulative_doses_taken—form a tight, strongly inter-correlated cluster ($r > 0.85$), confirming internal measurement consistency. Anthropometric features (weight, height) are orthogonal to all adherence measures ($r \sim 0$), indicating that body-size variation is independent of treatment compliance in this cohort. Retention features (last_month_attended, total_observed_months) show only weak positive associations with adherence ($r \approx 0.2$), suggesting that visit persistence and dose compli-

Figure 14*Temporal Lag Correlation of Percentage Adherence Across Consecutive Treatment Months*

ance capture distinct dimensions of patient engagement. Diagnostic variables such as `smear_microscopy_result` exhibit near-zero correlations with adherence and retention, consistent with their role as outcome indicators rather than behavioral predictors. Xpert MTB/RIF does not appear in the matrix because all observed values in the temporal subset were identical (zero variance), producing undefined Pearson coefficients.

Table 9
Lag-1 Temporal Correlation of Pct Adherence Across Consecutive Treatment Months (Justifying Time-Series Model Architecture)

Lag Pair	Pearson r	p-value	Significant?	N (pairs)
M0 → M1	0.9242	0.0000	Yes	300
M1 → M2	1.0	0.0000	Yes	285
M2 → M3	1.0	0.0000	Yes	275
M3 → M4	1.0	0.0000	Yes	271
M4 → M5	0.9999	0.0000	Yes	254
M5 → M6	1.0	0.0000	Yes	197
M6 → M7	0.6202	0.0000	Yes	80
M7 → M8	0.2066	0.6945	No	6

The month-to-month stability of adherence behavior in the temporal dataset provides empirical support for the choice of temporal model architecture. Early treatment months exhibited near-perfect lag-1 correlations (M0–M1: $r = 0.92$, $p < 0.001$; M1–M2: $r = 1.00$, $p < 0.001$), indicating that adherence is highly autocorrelated in the initial treatment phase. Correlation remained above $r = 0.99$ through M5–M6,

then declined at M6–M7 ($r = 0.62$, $p < 0.001$) before becoming non-significant at M7–M8 ($r = 0.21$, $p = 0.69$), where only six patient pairs remained available. This sustained autocorrelation confirms that adherence carries meaningful sequential structure, justifying the use of recurrent architectures (RNN, LSTM) that model temporal dependencies (Nayebi et al., 2022). The decline after M6 also aligns with the retention cascade, suggesting that dropout-compromised patients exhibit increasingly erratic adherence before loss to follow-up.

Discussion

This study investigated the use of machine learning models to predict treatment outcomes—Success or Failure—of PTB patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. Both temporal models, specifically the Bi-LSTM Baseline and Augmented variants, and tree-based models, including XGBoost, LightGBM, and Random Forest with SMOTE-ENN, were evaluated on a stratified 70/20/10 patient-level split, incorporating progressive temporal samples from treatment months M0 through M12. The test set performance revealed that while Bi-LSTM and XGBoost Baseline models achieved high overall accuracy (0.9048) and F1 scores (0.9500), their ability to detect treatment failures was limited, as indicated by zero specificity for the minority Failure class. In contrast, Random Forest with SMOTE-ENN demonstrated more balanced performance, achieving a recall of 0.7895 for treatment failure and a high PR-AUC (0.9419), indicating greater reliability in identifying patients at risk of adverse outcomes. These results highlight that tree-based models augmented with SMOTE-ENN are particularly effective at handling severe class imbalance, while temporal models provide strong predictive accuracy for the majority outcome.

The study contributes in several ways. By integrating temporal monitoring data with static baseline features, it demonstrates that patient-level longitudinal information improves predictive performance over static data alone. The application of SMOTE-ENN and advanced data augmentation techniques for Bi-LSTM effectively addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Furthermore, the findings suggest that predictive models, particularly Random Forest with SMOTE-ENN, have practical value for clinical decision support systems, enabling clinicians to identify patients at risk of treatment failure early and prioritize interventions accordingly.

Based on these findings, it is recommended that healthcare facilities and policymakers consider integrating machine learning models like Random Forest with SMOTE-ENN into existing TB monitoring programs to improve treatment outcomes and optimize resource allocation. Future research should focus on larger, multi-region datasets to enhance generalization, explore hybrid modeling approaches combining deep learning and tree-based methods, and investigate additional features such as social determinants of health and geographic risk factors. Additionally, enhancing model interpretability, particularly of Bi-LSTM attention outputs, could improve clinician trust and adoption.

Safety Scope and Interpretation Limits

The narrative explanation layer improves accessibility of model reasoning by translating contribution signals into clinician-readable text, but its outputs must be interpreted within clear safety boundaries. The explanation should be read as describing how factors increased or decreased the model-predicted risk, not as establishing causal mechanisms of disease progression. This distinction is important in high-stakes settings where generative outputs may otherwise be over-interpreted beyond their evidentiary basis (Vasey et al., 2022; World Health Organization, 2021).

Accordingly, the system is positioned as decision support: it summarizes model behavior while preserving clinician authority for diagnosis and treatment decisions. This framing is consistent with contemporary reporting and governance guidance for AI-assisted clinical prediction systems, where transparency, bounded use, and human oversight remain central requirements (Collins et al., 2024).

Acknowledgments

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Appendix A: iCreate Supplementary Material

Treatment Success and Failure Prediction using Machine Learning Models for Pulmonary Tuberculosis Patients under the DOH-CAR TB-DOTS Program

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Abstract

This study proposes a multi-layer analytical framework using machine learning to predict pulmonary tuberculosis (PTB) treatment success or failure in the Cordillera Administrative Region (CAR), Philippines. Utilizing demographic, clinical, and diagnostic data from 2015 to 2025, supervised models (Logistic Regression, Support Vector Machines, and Extreme Gradient Boosting) and temporal models (Recurrent Neural Networks and Long Short-Term Memory) are trained to identify risk factors. Shapley Additive Explanations (SHAP) ensure interpretability. The models will be integrated into a clinical decision support tool for early intervention. Aligning with the Sustainable Development Goals for good health and well-being, this predictive tool aims to enhance TB control efforts, optimize resource allocation, and improve patient management.

Keywords

Clinical decision support, machine learning, predictive modeling, pulmonary tuberculosis, treatment outcomes

1. Introduction

Pulmonary Tuberculosis (PTB) is a chronic infectious disease that continues to challenge public health due to its gradual progression and risk of treatment failure (Tobin & Tristram, 2024). In the Philippines, TB remains a major public health concern, with the country being one of the top contributors to the global TB burden (World Health Organization, 2022). In the Cordillera Administrative Region (CAR), the Department of Health (DOH) reported 437 confirmed TB cases in Baguio City alone in 2022, alongside an estimated 4,405 missing or unreported cases (Department of Health Republic of the Philippines, n.d.).

While multiple studies have examined risk factors such as sociodemographic characteristics, comorbidities, and medical history, findings remain fragmented as traditional statistical analyses often assume linear relationships and examine only a few factors simultaneously (Amante &

Abdosh, 2015). To address this gap, this study adopts an integrative machine learning framework to combine multiple dimensions of data in analyzing interrelated variables.

The specific research objectives of this study are as follows:

1. **Data Pre-processing & Feature Engineering:** To clean, pre-process, and digitalize PTB patient records, creating engineered features that ensure the dataset is optimized for statistical analysis and machine learning applications.
2. **Statistical Analysis & Profiling:** To perform descriptive and inferential statistics to profile the demographic, clinical, and treatment characteristics of the study population, identifying baseline variables that significantly influence treatment outcomes.
3. **AI Modeling & Comparative Analysis:** To develop, validate, and evaluate both non-temporal and temporal machine learning models for predicting PTB treatment outcomes, comparing the predictive performance of static baseline data against dynamic, time-dependent patient patterns throughout the course of therapy.
4. **Data Visualization & Clinical Interpretation:** To visualize patient risk levels, statistical trends, and predictive outputs from the AI models, providing clear insights to support clinical interpretation and the early identification of individuals at risk of poor outcomes.

2. Methodology

This study utilizes a retrospective dataset of PTB patients from the CAR covering the period from 2015 to 2025, consolidated by the DOH-CAR. Data undergoes a standardized preprocessing pipeline, including schema harmonization, error correction, and missing value imputation via Multivariate Imputation by Chained Equations, or MICE (Alruhaymi & Kim, 2021).

To predict treatment outcomes defined by the National Tuberculosis Program criteria, supervised models including Logistic Regression, Support Vector Machines (SVM), and Extreme Gradient Boosting (XGBoost) are utilized. Furthermore, temporal models using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) are employed to capture longitudinal treatment dynamics (Lu et al., 2022; Nayebi et al., 2022). Model performance is assessed using standard classification metrics, and Shapley Additive Explanations (SHAP) are applied to ensure clinical interpretability (Peng et al., 2024).

3. Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static (Random Forest, Gradient Boosting, LightGBM, XGBoost, Logistic Regression) and temporal models are compared, and feature importance analyses provide insights into the key factors influencing treatment outcomes. The discussion interprets these findings in

the context of clinical relevance and public health priorities, highlighting implications for early intervention, resource allocation, and strengthened TB management.

3.1 Dataset Characteristics and Class Distribution

The final dataset exhibited significant class imbalance, with 86.2% of cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, including four newly introduced geographic and facility-level variables and one derived clinical delay variable (Diagnosis_to_Treatment_days). Resampling using SMOTE combined with Edited Nearest Neighbors (SMOTE+ENN) was applied exclusively to the training set, increasing the minority (failure) class by 433% and achieving a near-balanced 0.70:1 ratio. This approach mitigated class bias while preserving realistic evaluation conditions in the test and validation sets.

3.2 Validation Set Performance and Generalization

Table 1: Test Set Model Performance

Model	AUC (Area Under the Curve)	Recall (Failure)	F1 (Failure)
Random Forest	0.7154	57.98%	0.3539
Gradient Boosting	0.7121	44.68%	0.3262
LightGBM	0.7119	48.40%	0.3467
XGBoost	0.7095	47.87%	0.3416

Among all evaluated models, Random Forest achieved the highest overall performance on the test set, with an AUC of 0.7154 and an F1-score of 0.3539 for the failure class. It also demonstrated balanced recall (57.98%) without excessive false positives. Gradient Boosting, LightGBM, and XGBoost achieved comparable AUC values (~0.71), though with slightly lower recall for treatment failure. Logistic Regression demonstrated the highest recall (77.13%); however, its precision was notably low (19.6%), resulting in over-alerting and reduced overall accuracy (~51%). These findings suggest that while Logistic Regression aggressively identifies high-risk patients, it lacks specificity, potentially burdening healthcare systems with excessive false alarms.

On the 10% held-out validation set, Random Forest maintained superior performance, achieving a validation AUC of 0.7380 and recall of 52.13% for treatment failure. Notably, all models demonstrated positive AUC gains from test to validation sets, indicating strong generalization and absence of overfitting. The consistency of Random Forest across both evaluation phases confirms its robustness and suitability for deployment within a clinical decision support framework.

3.3 Validation Set Performance and Generalization

Table 2: Test Set Performance Metrics of Temporal Models for TB Treatment Outcome Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

Table 1 presents the performance metrics of all temporal models evaluated on the held-out test set. Both Bi-LSTM variants and XGBoost Baseline achieved the highest overall accuracy (0.9048) and F1 score (0.9500), reflecting their ability to correctly predict the dominant Success class. Precision values were similarly high (0.9048), indicating few false positives for the majority class. However, recall for the minority Failure class reached 1.0 for these models, while specificity remained 0.0, demonstrating that no actual Failure cases were correctly identified in the small test set. Tree-based models augmented with SMOTE-ENN, particularly Random Forest, showed a more balanced performance. Random Forest achieved slightly lower overall accuracy (0.7143) but higher recall for the Failure class (0.7895) and competitive PR-AUC (0.9419), indicating better identification of high-risk patients. ROC-AUC scores further illustrate differences in probability calibration, with XGBoost Baseline achieving the highest ROC-AUC (0.5789), while the Bi-LSTM Augmented variant performed lower (0.3684), likely due to overfitting on limited Failure samples. Overall, these

results suggest that while Bi-LSTM and XGBoost excel at predicting the majority outcome, Random Forest with SMOTE-ENN offers the most effective detection of treatment failure, making it more suitable for clinical decision support applications.

4. Conclusions and Recommendations

This study investigated the use of machine learning models to predict treatment outcomes—Success or Failure—of pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. Both temporal models, specifically the Bi-LSTM Baseline and Augmented variants, and tree-based models, including XGBoost, LightGBM, and Random Forest with SMOTE-ENN, were evaluated on a stratified 70/20/10 patient-level split, incorporating progressive temporal samples from treatment months M0 through M12. The test set performance revealed that while Bi-LSTM and XGBoost Baseline models achieved high overall accuracy (0.9048) and F1 scores (0.9500), their ability to detect treatment failures was limited, as indicated by zero specificity for the minority Failure class. In contrast, Random Forest with SMOTE-ENN demonstrated more balanced performance, achieving a recall of 0.7895 for treatment failure and a high PR-AUC (0.9419), indicating greater reliability in identifying patients at risk of adverse outcomes. These results highlight that tree-based models augmented with SMOTE-ENN are particularly effective at handling severe class imbalance, while temporal models provide strong predictive accuracy for the majority outcome.

The study contributes in several ways. By integrating temporal monitoring data with static baseline features, it demonstrates that patient-level longitudinal information improves predictive performance over static data alone. The application of SMOTE-ENN and advanced data augmentation techniques for Bi-LSTM effectively addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Furthermore, the findings suggest that predictive models, particularly Random Forest with SMOTE-ENN, have practical value for clinical decision support systems, enabling clinicians to identify patients at risk of treatment failure early and prioritize interventions accordingly.

Based on these findings, it is recommended that healthcare facilities and policymakers consider integrating machine learning models like Random Forest with SMOTE-ENN into existing TB monitoring programs to improve treatment outcomes and optimize resource allocation. Future research should focus on larger, multi-region datasets to enhance generalization, explore hybrid modeling approaches combining deep learning and tree-based methods, and investigate additional features such as social determinants of health and geographic risk factors. Additionally, enhancing model interpretability, particularly of Bi-LSTM attention outputs, could improve clinician trust and adoption.

Overall, this study aligns with Sustainable Development Goal 3 – Good Health and Well-Being by providing a data-driven approach to improving TB treatment outcomes. By accurately identifying patients at risk of treatment failure, these predictive models have the potential to reduce TB-related morbidity and mortality, inform targeted public health interventions, and contribute to more efficient and effective TB care within the Cordillera Administrative Region.

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